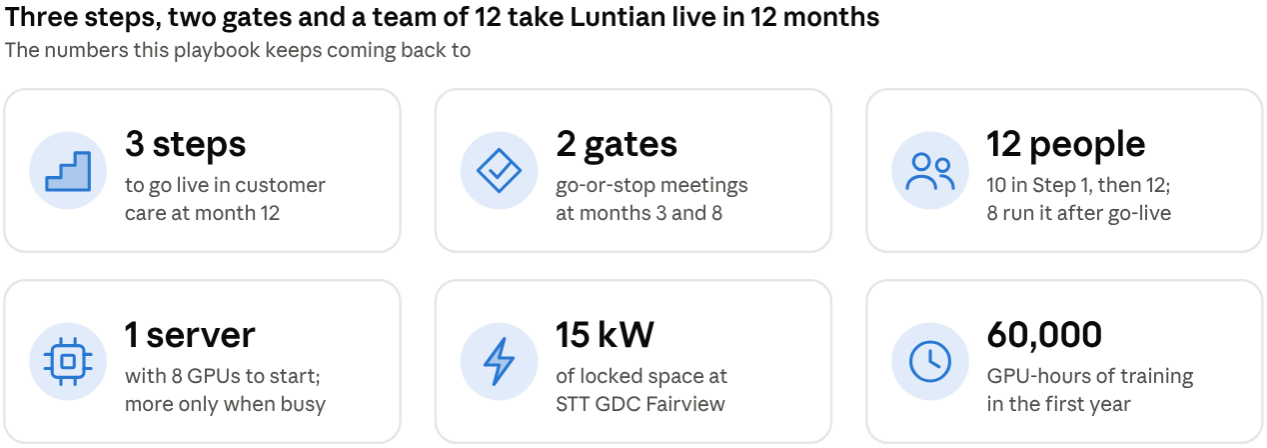
Project Luntian: How We Do It, Step by Step

Independent proposal, revised November 2026

This is the complete playbook for Project Luntian. For every stage it shows what must be in place first, what to do and when, and who does it. It also shows what each stage needs, what Globe has at the end, and the checks that must pass before more money is released.

It goes with the proposal, Project Luntian: The Proposal in Plain Words. It is an independent proposal for Globe Telecom, not an official Globe document.



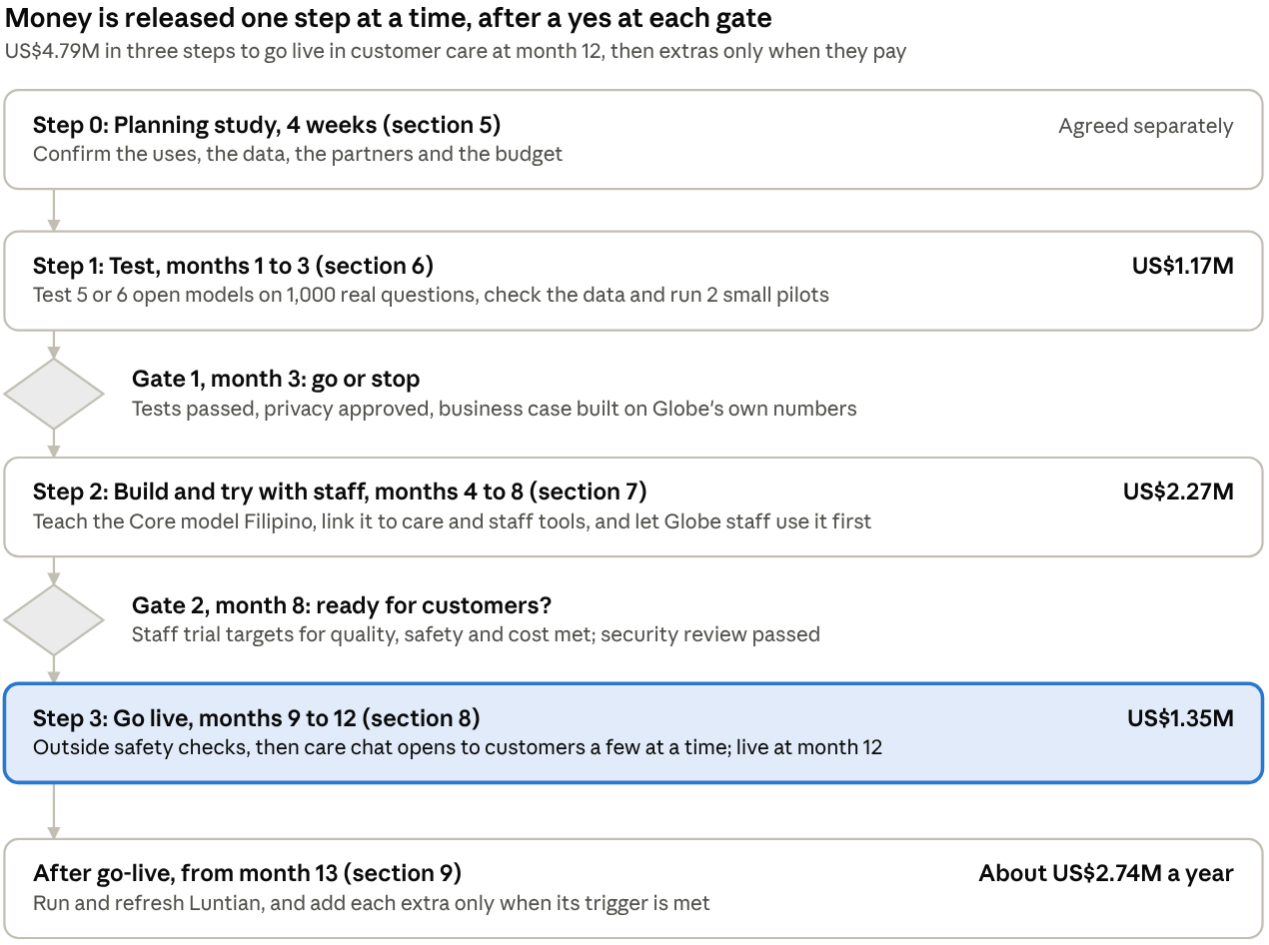
The plan in six numbers

Luntian starts small. Globe spends US$1.17M on a 3-month test first, about US$4.8M (₱299M) in total to go live in customer care at month 12, and about US$2.7M a year to run it after that. Anything more is added only when its trigger is met and it pays. Everything below explains how, stage by stage.

For a quick look first, the one-page website Project Luntian tells the same story as a comic in about five minutes.

1. How to use this playbook

The playbook follows the work in the order it happens: what to set up before Step 0, then each stage from the planning study to life after go-live, then the parts that run through every stage.



The stages of Project Luntian, with two gates

Every stage is written the same way:

1. **Goal, time, money and team**, in one line.
2. **What must be in place first.** A checklist of everything the stage needs before day one. If an item is not ticked, the stage waits.
3. **The work, activity by activity.** Each activity has a number, a lead, its timing, what it needs first, the tasks and what it produces.
4. **Milestones.** The few dates that show the stage is on track.
5. **What it needs and what it costs.** People, computing, data and tools, and the cost of each line.
6. **What Globe has at the end.** The things Globe keeps, whatever happens next.
7. **The gate.** The checklist leaders use before they release the next amount of money, and what happens if the answer is no.

Where to find things

Part	Sections	What it covers
The basics	1 to 3	How to read this, what gets built, and who owns each area of work
The stages	4 to 9	Before Step 0, Step 0, Steps 1 to 3, and life after go-live, including each add-on
How each part works	10 to 16	Team, training, data, data centre, buying, safety and green rules
Money and checks	17 to 19	Budget and payback, the numbers tracked at each gate, and risks
Reference	20 and 21	Glossary and sources

Checklists you can tick

- Before Step 0: section 4
- Before each stage, and at each gate: sections 5 to 8
- Go-live readiness: section 8
- Before any add-on starts: section 9
- Before opening a role: section 10
- Before each training run, and before each model release: section 11
- Before using a new data source: section 12
- The secure zone: section 13
- Before signing a supplier contract: section 14
- What to do if something goes wrong: section 15

Tick them in this doc, or copy them into Globe's own project tools.

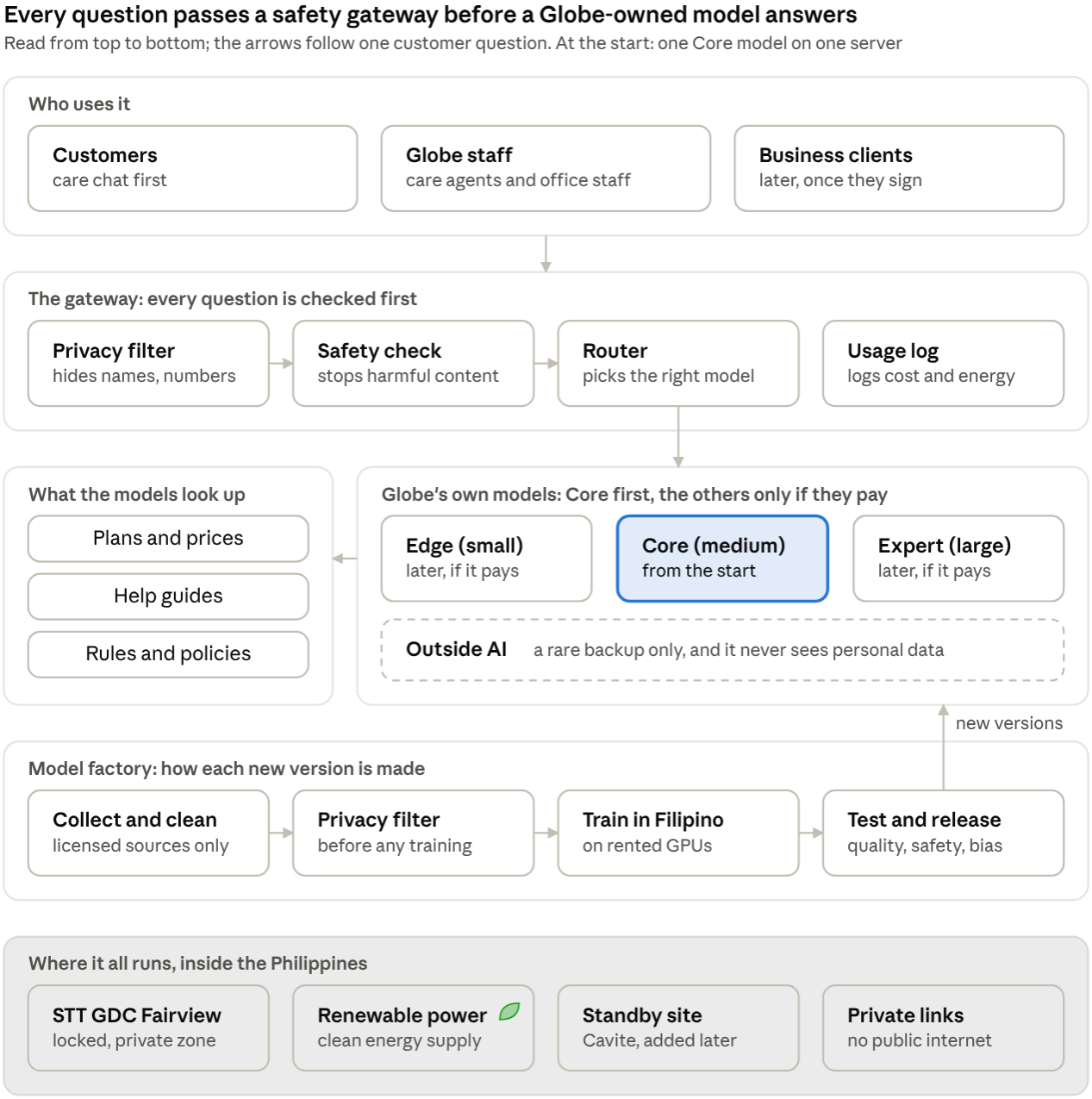
Numbering. Activities are numbered by step: 1.1 to 1.11 in Step 1, 2.1 to 2.10 in Step 2, 3.1 to 3.8 in Step 3, and R.1 to R.5 for running the service after go-live. The add-ons are A1 to A7. The same numbers are used in the weekly plans and the gate packs.

Costs are upper-side estimates in US dollars at ₱62.5 per US$, with a 15% margin for surprises on every line. They match the proposal.

2. What we are building

Luntian starts as one Globe-owned AI model, the medium-sized Core model, behind a gateway that protects data and a router that sends each question to the right place. It runs on one server in a locked space that Globe rents in the Philippines, on renewable

power. A small model and a large model are added later only if they pay.



What we are building: the whole system in one picture

Everyone comes in through one gateway. It hides personal details, blocks harmful content and sends each question to Globe's own model. The model factory at the bottom is where the training work in sections 6 to 8 and 11 happens, and section 13 explains the data centre.

The models

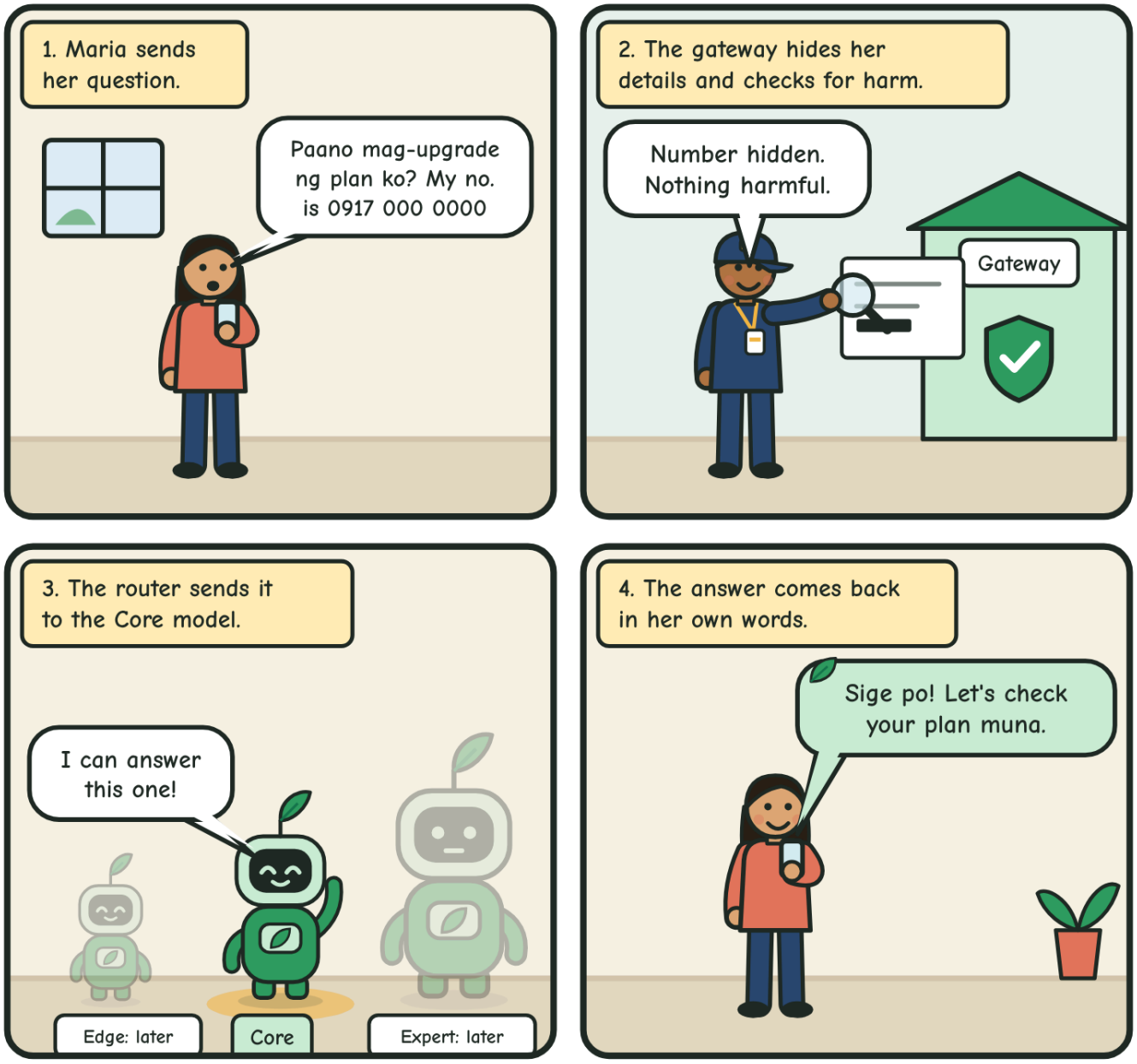
Model	Size	Where it runs	What it does	When
Core	About 26B to 30B in total, with only 3B to 4B used for each word (a design called mixture of experts)	One 8-GPU server in Globe's locked space at STT GDC Philippines	Care chat replies, help for agents, and the staff assistant	From the start: built in Step 2, live at month 12
Outside AI service	A contracted model that Globe does not own	The supplier's cloud	Rare, very hard questions, sent only with personal details removed	A safety net from Step 1
Edge, an add-on	2B to 4B parameters	Globe's network and, later, the GlobeOne app	Scam warnings in Filipino and quick sorting of requests	Only if add-on A6 is approved (section 9)
Expert, an add-on	27B to 32B, all used for each word	A second server in the same locked space	Long documents and the hardest questions	Only if add-on A7 is approved (section 9)

What Globe owns at the end

- The trained model, within the licence of the open model it starts from.
- The Filipino datasets, the test questions and every test result.
- The platform: gateway, router, safety tools and dashboards.
- Any work built with partners, which the contracts assign to Globe.

Who uses it first: Globe's care agents and customers in care chat, and Globe staff through the staff assistant. Voice, business clients and other channels come only through the add-ons in section 9, each when its trigger is met.

The same journey as a comic, following one customer question:



3. The workstreams and who owns them

The work runs in eight workstreams, plus business clients if that add-on starts. Each has one owner, so every task in this playbook has a home and a person who answers for it.

Workstream	What it covers	Owner	Main helpers
Programme and money	Plan, budget, suppliers, risks, gates and reports to the sponsor	Programme lead	Finance, procurement, legal
Product and adoption	Which uses come first, the targets, the pilots and the rollout	Product owner	Care operations, business units, HR

Workstream	What it covers	Owner	Main helpers
Data	Finding, licensing, cleaning and masking data; worked examples and test questions	Data engineer	Filipino linguists, legal, data privacy office, universities
Models	Choosing the open model, training it and testing its quality	Research and model lead	Model engineers, AI Singapore (Singapore's national AI programme)
Platform and data centre	Computing, gateway and router, monitoring, links to Globe systems, the secure zone	MLOps and platform engineer	Network team, cybersecurity, STT GDC Philippines
Safety and testing	Test sets, red teaming, the safety filters and release sign-off	Evaluation and safety lead	Ethics board, linguists
Privacy and governance	Privacy assessments, the privacy regulator, the ethics board and audits	Privacy and AI governance lead	Data protection officer, legal, cybersecurity
Green and cost	Energy, carbon and cost per answer	MLOps and platform engineer	Sustainability team, finance
Business clients, add-on A2	The offer, prices and sales to banks, BPOs, hospitals and government agencies	Globe Business	Product owner, legal

How the workstreams meet. Each squad holds a short daily stand-up. Workstream owners meet the programme lead once a week to clear blockers, and the steering committee meets every two weeks. Section 10 has the full meeting rhythm and who decides what.

4. Before Step 0: what must be in place

Step 0 can start once seven things are in place. None of them needs money for computers, new hires or data leaving Globe.

- [ ] **A senior sponsor.** A Globe chief officer agrees to sponsor the planning study and give it about an hour a week for four weeks.

- [ ] **A day-to-day contact.** One named person at Globe who books meetings, finds documents and opens doors.
- [ ] **A programme lead for Step 0.** A Globe manager or a contracted lead, named by the sponsor, who runs the four weeks.
- [ ] **A confidentiality agreement.** Signed between Globe and everyone working on the study, so internal numbers and sample data can be shared safely.
- [ ] **Time from five leads.** The leads of customer care, marketing, Globe Business, IT and the data privacy office each give about two hours a week.
- [ ] **Basic numbers.** Read access to care contact volumes and costs, current spending on outside AI tools, and the list of Globe's main data systems.
- [ ] **A one-page scope note.** Signed by the sponsor: the questions Step 0 must answer, who attends the final meeting, and its date.

What Step 0 does not need: no computers to rent or buy, no customer data leaving Globe, no hiring and no partner contracts. Any sample data used in Step 0 is checked first and holds no personal details.

5. Step 0: Planning study (4 weeks)

Goal: agree what Luntian should do first, and get Step 1 ready to approve. **Time:** 4 weeks. **Team:** the programme lead, with Globe's care, marketing, business, IT and privacy leads.

What must be in place first: the seven items in section 4.

The work, week by week

Week 1: agree the goal

- [ ] Kick-off with the sponsor: confirm the questions, the scope and the date of the decision meeting
- [ ] Interview the care, marketing, Globe Business, IT, network and privacy leads, for about an hour each
- [ ] Collect the 10 most promising uses, and how each would be measured
- [ ] Write down what success looks like for Step 1, in numbers where possible

Week 2: check the data

- [ ] List the data Globe could use: care chats and call transcripts, the knowledge base, network tickets and app reviews
- [ ] For each source, note the owner, rough volume, languages and whether it holds personal data
- [ ] Ask the data privacy office about the legal basis for each source, and any limits

- [ ] Mark SIM registration records as excluded from the start, as the SIM Registration Act requires

Week 3: shortlist uses, partners and models

- [ ] Pick the top uses for care chat and the staff assistant: highest value, lowest risk, and data that is already available
- [ ] Shortlist partners: AI Singapore, Philippine universities, publishers and broadcasters (none approached yet)
- [ ] Run quick tests of two or three open models on 50 to 100 sample questions with no personal data
- [ ] Ask STT GDC Philippines for an indicative offer for a small locked space of about 15 kW, for one 8-GPU server from month 4

Week 4: plan Step 1

- [ ] Write the Step 1 plan: the 11 activities, their weeks, the team and the US$1.17M budget
- [ ] Draft the Gate 1 targets: test score, privacy, the model licence, the business case and the small locked space
- [ ] Name the people for the 10 Step 1 roles, from AIDE (Globe's in-house AI group), Globe IT, partners or contractors, and draft job descriptions for the permanent roles
- [ ] List the approvals Step 1 needs, with owners and dates: budget, a security sign-off for rented computing, and data access
- [ ] Present the plan to the sponsor and set the date for the Step 1 decision

What it needs and what Globe has at the end

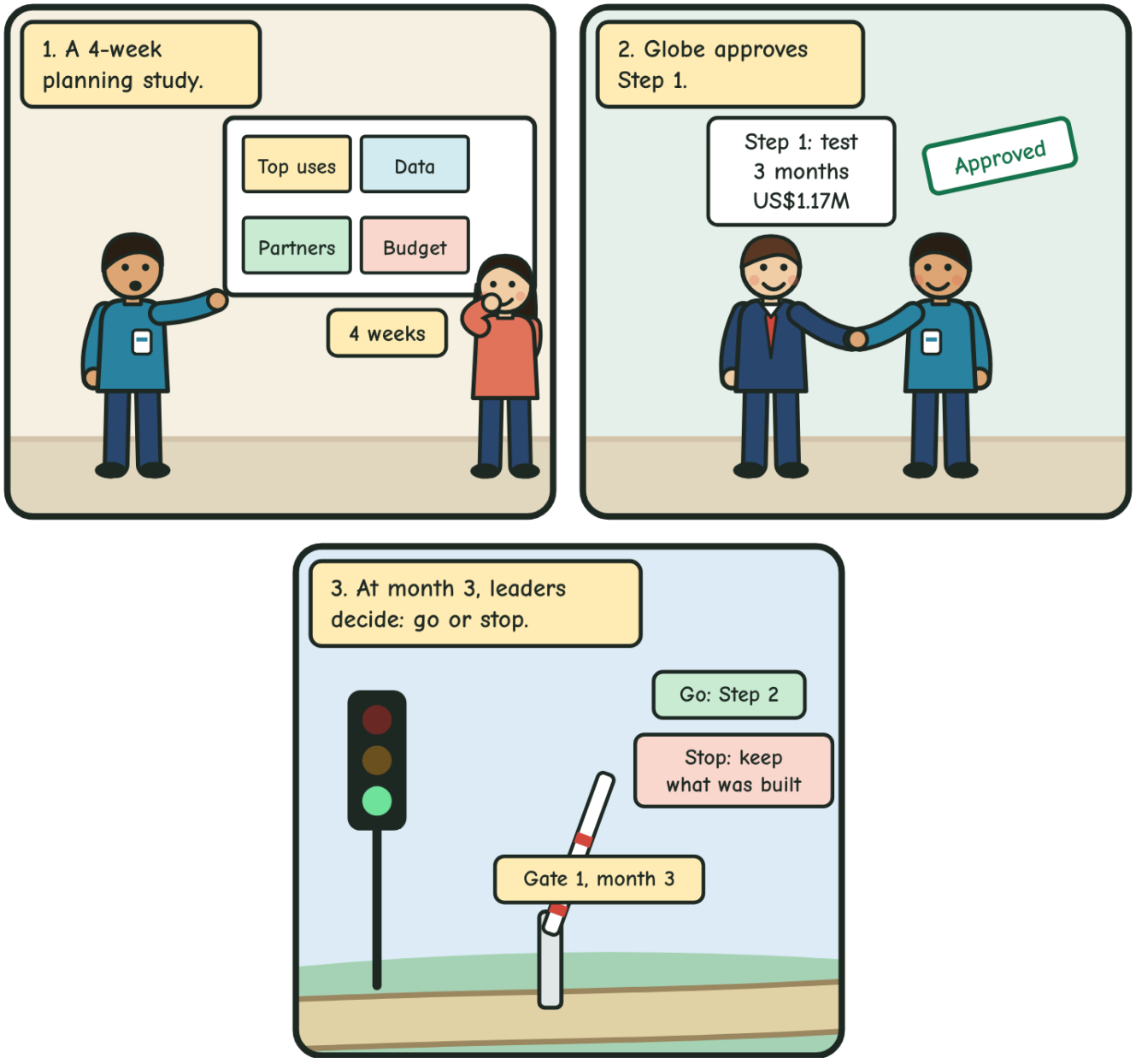
What it needs from Globe: a senior sponsor and about two hours a week from each of the five leads. It also needs sample data with no personal details under the confidentiality agreement, and the basic numbers listed in section 4. Step 0 needs no money for computing or data. Its cost, mainly the time of the people involved, is agreed separately and is not in the US$4.79M.

What Globe has at the end:

- A named executive sponsor for the programme
- The top uses for care chat and the staff assistant, each with a target and an owner
- A first data and privacy check: each source, its owner and whether it can be used
- A shortlist of partners and open models
- A Step 1 plan, team and budget, ready to approve
- A list of the approvals Step 1 needs, with owners and dates

The decision at the end of Step 0

The sponsor takes the Step 1 plan to Globe's leaders, who approve Step 1 (US$1.17M for 3 months) or stop. If they stop, Globe keeps the list of uses, the data list and the plan, and nothing more is spent.



6. Step 1: Test (months 1 to 3)

Goal: find out whether an adapted open model can do Globe's Filipino work well, safely and at a good price, before more money is spent. **Time:** 12 weeks. **Money:** US$1.17M (₱73M). **Team:** 10 people.

The 10 are the programme lead, product owner, research and model lead, two model-training engineers, a data engineer, a Filipino linguist, an MLOps and platform engineer, an evaluation and safety lead, and a privacy and AI governance lead. Up to half can come

from AIDE and Globe IT, and partners or contractors fill the rest, because outside hiring takes longer than this step.

What must be in place first

- [ ] Globe's leaders have approved the Step 1 budget of US$1.17M, and finance has opened a cost centre
- [ ] The executive sponsor is named, and the programme lead and product owner are appointed
- [ ] All 10 Step 1 roles are filled from AIDE, Globe IT, partners or contractors, with start dates in week 1 or 2
- [ ] The draft Gate 1 targets from Step 0 are ready for the sponsor to confirm in week 2
- [ ] Globe's cybersecurity team has approved rented cloud GPUs for work with no personal data, and a test server rented in the Philippines for the pilots
- [ ] An account with a pay-per-use AI service is approved for comparison tests and as a back-up in the pilots, for questions with no personal data
- [ ] Each data source on the Step 0 list has a named owner who will answer the data audit
- [ ] The team has laptops, accounts and building access from day one
- [ ] Legal has confidentiality and partner agreement templates ready

The work, activity by activity

1.1 Mobilise. Lead: programme lead. Weeks 1 to 2.

- **Needs first:** Step 1 approved, the sponsor named, and the Step 0 plan.
- **Do:**
  - [ ] Bring the team together and hold a kick-off with the sponsor
  - [ ] Write the charter: goal, scope, roles and how decisions are made
  - [ ] Agree the KPIs and Gate 1 targets with the sponsor (section 18)
  - [ ] Record today's numbers for handling time, containment, satisfaction and cost per contact, so the pilots have a baseline
  - [ ] Set up the meeting rhythm in section 10, and open the risk log and the decision log
- **Output:** a signed charter, a KPI sheet with baselines and Gate 1 targets, and working risk and decision logs.

1.2 Data audit. Lead: data engineer, with the privacy lead. Weeks 1 to 6.

- **Needs first:** data owners named, read access to the list of systems, and the confidentiality agreement signed by the team.
- **Do:**

- [ ] List every source: care chats, call transcripts, the knowledge base, network tickets and app reviews
- [ ] For each, record the owner, legal basis, volume, languages and quality
- [ ] Check quality on a small sample, masked first
- [ ] Mark the sources that hold personal data, and exclude SIM registration records

- **Output:** the data inventory, agreed by each data owner.

1.3 Privacy impact assessment and ethics board. Lead: privacy lead. Weeks 1 to 6, ethics board by week 10.

- **Needs first:** the draft data inventory from 1.2, Globe's assessment template, and time from Globe's data protection officer.
- **Do:**
  - [ ] By week 2, give an early privacy decision on two narrow uses: a masked sample of care questions for the test sets, and knowledge-base articles for the staff pilot
  - [ ] Check each source against the Data Privacy Act and the privacy regulator's 2024 advisory on AI
  - [ ] Decide what may be used, for what purpose and with what masking, and list the customer notices that need updating
  - [ ] Sign the full assessment by week 6, before any customer question reaches a pilot
  - [ ] Set up an ethics board of three to five people, including outside members, with written terms of reference
- **Output:** an early privacy decision (week 2), a signed privacy impact assessment and a list of approved sources (week 6), and an ethics board that has met once.

1.4 Golden test sets. Lead: linguist, with the product owner. Weeks 2 to 6.

- **Needs first:** the early privacy decision from 1.3, about 2,000 real customer questions masked and checked by the data engineer, and care experts free for about two hours a week.
- **Do:**
  - [ ] Pick about 1,000 questions that cover the top uses in care chat and the staff assistant
  - [ ] Rewrite each question so it holds no personal details, and write the agreed answer in Filipino, Taglish and Cebuano
  - [ ] Add hard and tricky cases
  - [ ] Split them into a working set and a locked set used only at gates
  - [ ] Agree how answers are scored
- **Output:** golden test sets with no personal data (the first 500 by week 4, all 1,000 by week 6), kept apart from any training data.

1.5 Model bake-off. Lead: research and model lead. Weeks 2 to 8.

- **Needs first:** rented GPU time from 1.7, five or six open models whose licences allow commercial use, and the first 500 golden questions from week 4.
- **Do:**
  - [ ] Run each model on public Filipino tests (FilBench and SEA-HELM), then on the working golden set
  - [ ] Compare them with the outside AI that Globe can rent today, on the same questions
  - [ ] Measure answer quality, speed, cost and energy per 1,000 answers
  - [ ] Try light tuning on the top two
  - [ ] Send the shortlisted licences to legal
- **Output:** a scorecard and a recommended base model for the Core model, with a back-up choice.

1.6 Quick-win pilots. Lead: product owner, with the MLOps and platform engineer. Weeks 3 to 10.

- **Needs first:** the test server and the masking step from 1.7, knowledge-base articles cleared in 1.3, about 50 staff and 20 to 30 care agents willing to try it, and the baselines from 1.1.
- **Do:**
  - [ ] Set up an open model with document search over the knowledge base
  - [ ] From week 4, run the staff pilot: an internal assistant that uses knowledge-base content only
  - [ ] From week 6, once the privacy assessment is signed, run the care pilot: suggested replies for care agents, who check every one
  - [ ] Mask customer details before any question reaches a model
  - [ ] Collect usage, quality ratings and time saved every week, against the baselines
- **Output:** pilot results measured against the week 2 KPIs.

1.7 MLOps set-up. Lead: MLOps and platform engineer. Weeks 2 to 10.

- **Needs first:** the security sign-off for rented computing, a cloud account with a budget, and a short security review of each tool.
- **Do:**
  - [ ] Rent GPU time for the bake-off, and a smaller test server in the Philippines for the pilots
  - [ ] Set up a masking step for names, numbers and addresses, and test it on samples with the privacy lead before the pilots use it

  - [ ] Set up experiment tracking, a model registry, a search index and monitoring
  - [ ] Keep test and pilot environments apart
  - [ ] Switch on cost and energy metering from the start
- **Output:** a working platform for the bake-off and the pilots, with tested masking.

1.8 Partner and data terms. Lead: programme lead, with legal. Weeks 2 to 10.

- **Needs first:** the partner shortlist from Step 0, legal's templates, and the sponsor's agreement on what Globe offers each partner.
- **Do:**
  - [ ] By week 4, agree a short advisory agreement with AI Singapore, Singapore's national AI programme, for Step 1
  - [ ] Agree term sheets for joint work from Step 2 with AI Singapore and Philippine universities, giving Globe ownership of joint work
  - [ ] Agree licence terms in principle with publishers and broadcasters, with prices, ready to sign after Gate 1
- **Output:** an advisory agreement in place, and partner term sheets and licence terms ready to sign after Gate 1.

1.9 Space and capacity plan. Lead: MLOps and platform engineer, with cybersecurity and procurement. Weeks 4 to 10.

- **Needs first:** rough sizes from the bake-off, a contact at STT GDC Philippines, and quotes from at least three suppliers.
- **Do:**
  - [ ] Size the computing: about 40,000 GPU-hours of training in Step 2 and 10,000 in Step 3, and one 8-GPU server to answer staff and customers
  - [ ] Agree a small locked space of about 15 kW at STT Fairview for that one server, with room to add more servers one at a time later
  - [ ] Agree the secure zone rules with cybersecurity and STT GDC
  - [ ] Subject to Gate 1, agree the space contract with STT GDC and order its private links with the network team
  - [ ] Subject to Gate 1, book the Step 2 training GPU time, including GPUs in the Philippines for tuning on Globe data, and the rented 8-GPU server from month 4
  - [ ] Get quotes both for renting the server and for buying one, for the decision at month 12 (section 14)
  - [ ] Set the energy targets
- **Output:** a capacity plan, an agreed small space with secure zone rules, and Step 2 computing ready to confirm at Gate 1.

1.10 Business case. Lead: product owner, with finance. Weeks 8 to 12.

- **Needs first:** Globe's own volumes and costs (contacts, handling times, cost per contact, spending on outside AI), the pilot results from 1.6, and a finance partner.
- **Do:**
  - [ ] Size the value of care chat and the staff assistant with Globe's own numbers, in place of the planning assumptions in section 17
  - [ ] Name an owner for each source of value
  - [ ] Work out the payback for the three uses in section 17: care and staff only, plus 10 business clients, and plus 20
  - [ ] Note any business clients who show interest, for add-on A2 later; no letters of intent are needed at this gate
- **Output:** a business case signed off by finance.

1.11 Gate 1 pack. Lead: programme lead. Weeks 11 to 12.

- **Needs first:** the outputs of 1.2 to 1.10.
- **Do:**
  - [ ] Run the locked golden sets on the top two models
  - [ ] Put the evidence for each Gate 1 item in one short pack
  - [ ] Send it to the steering committee a week before the meeting
  - [ ] Hold the Gate 1 meeting
- **Output:** a go or stop decision, written in the decision log.

Milestones

Week	What is done
2	Charter, KPIs, baselines and Gate 1 targets signed; early privacy decision
4	First 500 test questions; staff pilot running; advisory agreement with AI Singapore
6	Data inventory and privacy assessment signed; all 1,000 test questions done; care pilot starts; check-in with the sponsor
8	Bake-off scorecard and shortlist; licences with legal
10	Pilot results; term sheets and licence terms agreed; small locked space and Step 2 computing ready to confirm; secure zone rules agreed; ethics board has met

Week	What is done
12	Gate 1 meeting

What it needs and what it costs

Needs: about 10,000 hours of rented GPU time, a smaller test server rented in the Philippines for the 3 months, and a pay-per-use AI service for comparison tests. It also needs tools to track and test the models, and time from Globe's privacy, legal, security, care and finance teams. It needs no data centre space of its own.

Cost line	What it pays for	US$k
People	10 people for 3 months: 30 person-months at US$13,750	413
Training computing	About 10,000 GPU-hours at US$6.20 for the bake-off and small trials	62
Serving and outside AI	A smaller test server rented in the Philippines (60) and outside AI for comparison tests (40)	100
Data	Data audit and privacy check (46), golden test sets (60), legal work on licences (40)	146
Platform	MLOps tools (120), security review (40), developer set-up (40)	200
Governance	Privacy impact assessment and legal (50), licence review (20), setting up the ethics board (30)	100
Margin for surprises	15% of the lines above	153
Total		**1,174 (₱73M)**

What Globe has at the end

- The data inventory and the signed privacy impact assessment
- 1,000 golden test questions in Filipino, Taglish and Cebuano, owned by Globe
- A scorecard of five or six open models, and a recommended base model with a back-up
- Results from two small pilots, measured against baselines and agreed targets
- A working platform for tracking and testing models, with tested masking
- Partner term sheets, a capacity plan, an agreed small space with secure zone rules, and a business case

Gate 1: go or stop (end of month 3)

Every item must be true to release Step 2 (US$2.27M). The evidence for each goes in the Gate 1 pack.

- [ ] The best open model beats today's AI on Globe's test questions in Filipino and Taglish, and meets its target (evidence: locked golden set results; confirmed by the research lead and the safety lead)
- [ ] The privacy impact assessment approves the data sources and how they are used (evidence: the signed assessment; confirmed by the privacy lead and the data protection officer)
- [ ] Legal has approved the base model's licence (evidence: legal's written note)
- [ ] The business case, built on Globe's own numbers, shows a payback the sponsor accepts, with a named owner for each source of value (evidence: the business case; confirmed by finance)
- [ ] A small locked space of about 15 kW is agreed with STT GDC Philippines, and the Step 2 computing is ready to book (evidence: the offer and the bookings; confirmed by the MLOps and platform engineer)

Also decided at Gate 1: which partners to sign, and whether the two Step 2 roles can be filled from Globe staff. Cebuano is tested at Gate 2, once more Cebuano data has been added.

If the answer is no: Globe stops, having spent at most US$1.17M. It keeps the test questions, the data audit, the tools and the lessons, and carries on with rented AI.

7. Step 2: Build and try with staff (months 4 to 8)

Goal: turn the chosen open model into Globe's Core model in Filipino, connect it to Globe's care and staff tools, and let Globe staff use it before any customer does. **Time:** 5 months. **Money:** US$2.27M (₱142M). **Team:** 12 people.

Two people join: a platform engineer, for the links to Globe's systems, and a second Filipino linguist, for the worked examples. Permanent hires for Step 1 roles that contractors held also join in month 4 (section 10).

What must be in place first

- [ ] Gate 1 is passed and the Step 2 budget of US$2.27M is released
- [ ] The two new roles were opened in week 6 of Step 1, with offers that depend on Gate 1, and have start dates in month 4
- [ ] Partner agreements with AI Singapore and at least one Philippine university are ready to sign, with joint work assigned to Globe

- [ ] Data licences with publishers and broadcasters are agreed in principle, with prices
- [ ] Training GPU time is booked: about 30,000 GPU-hours to teach the Core model Filipino, and about 10,000 more for tuning on Globe data in the Philippines and for trials
- [ ] The small locked space at STT GDC is contracted under the zone rules agreed in 1.9, and the rented 8-GPU server is ready to install there in month 4
- [ ] Private links from the space to Globe's care and staff test systems are ordered, to be live by month 6
- [ ] The privacy impact assessment covers training on Globe data, and customer notices are updated where needed
- [ ] The base model and its back-up are confirmed, and legal has approved both licences
- [ ] Care operations has named a lead for the staff trial and agreed its targets

The work, activity by activity

2.1 Sign data deals. Lead: programme lead, with legal. Months 4 to 5.

- **Needs first:** the licence terms agreed in 1.8, the US$250k budget for licensed text, and legal's licence checklist.
- **Do:**
  - [ ] Sign licences for Filipino news, books and broadcast transcripts
  - [ ] Check that each licence allows AI training and leaves Globe the rights to the trained model
  - [ ] Log each licence and its limits in the licence register
- **Output:** signed licences and a licence register.

2.2 Build the Filipino corpus. Lead: data engineer. Months 4 to 6.

- **Needs first:** signed licences from 2.1, and the approved source list from 1.3.
- **Do:**
  - [ ] Gather several billion tokens of licensed and public Filipino, Taglish, Cebuano and English text
  - [ ] Remove duplicates, spam, machine-translated text and offensive content, and balance the languages
  - [ ] Mask any names, numbers and addresses that remain
  - [ ] Keep Globe's own text (masked care transcripts and knowledge-base articles) in a separate set, stored in the Philippines and used only for tuning
  - [ ] Record where every record came from
  - [ ] Release a first version for training by month 5, and the full corpus by month 6
- **Output:** the training corpus of licensed and public text, a separate Globe set for tuning, and a short data sheet for each.

2.3 Write worked examples. Lead: linguists. Months 4 to 7.

- **Needs first:** a style guide for Globe's tone agreed with care and marketing, annotators contracted with universities, and a review process with two checks per example.
- **Do:**
  - [ ] Write and check thousands of worked examples for care chat and staff tasks
  - [ ] Include examples of what the AI must refuse
  - [ ] Where examples are drafted by AI to save time (synthetic data), check every one by hand
  - [ ] Keep real customer data out
- **Output:** teaching data, owned by Globe.

2.4 Teach the Core model Filipino. Lead: research and model lead. Months 5 to 6.

- **Needs first:** the first corpus release from 2.2, the booked GPU time, the base model confirmed at Gate 1, and experiment tracking.
- **Do:**
  - [ ] Continue training the Core model on the corpus, about 30,000 GPU-hours
  - [ ] Move to the full corpus when it is ready in month 6
  - [ ] Check progress on the working golden set every week, and keep the best version
- **Output:** a Core model that handles Filipino, Taglish and Cebuano well.

2.5 Teach Globe's tasks and make it safe. Lead: model engineers, with the evaluation and safety lead. Months 6 to 7.

- **Needs first:** the adapted model from 2.4, the worked examples from 2.3, a list of harms to test agreed with the ethics board, and booked GPUs in the Philippines, because this step uses Globe data.
- **Do:**
  - [ ] Fine-tune the Core model on the worked examples, in the tone Globe wants, on about 5,000 GPU-hours in the Philippines
  - [ ] Safety-tune it to refuse harmful requests
  - [ ] Run first red-team tests in Filipino, Taglish and Cebuano, then fix and test again, using about 5,000 GPU-hours for trials and reruns
  - [ ] Give an interim sign-off for staff use in month 7, with a person checking answers
- **Output:** a task-ready Core model by month 7, and an interim safety sign-off.

2.6 Build the gateway and router. Lead: MLOps and platform engineer. Months 4 to 7.

- **Needs first:** the platform from 1.7, a security design approved by cybersecurity, and the masking rules from the privacy lead.
- **Do:**

  - [ ] Build the front door that masks personal data and applies policy checks
  - [ ] Send each question to the Core model, or to the outside AI for the rare, very hard question, with personal details removed
  - [ ] Meter cost and energy, and add rate limits and spending caps
  - [ ] Log every request
  - [ ] Have Globe's cybersecurity team review the gateway, router and links before the staff trial
- **Output:** the gateway and router, working in test by month 7.

2.7 Install the server and connect Globe's tools. Lead: platform engineer, with the product owner. Months 4 to 7.

- **Needs first:** the space contract and links agreed in 1.9, the rented 8-GPU server, energy meters, access to the test versions of the contact-centre, knowledge-base and staff systems, and time from their owners.
- **Do:**
  - [ ] Install the rented 8-GPU server in the locked space at STT GDC in month 4, and meter its energy from day one
  - [ ] Use it for tuning tests on Globe data, then to answer staff from month 7
  - [ ] Plug Luntian into the tools agents and staff already use, so suggested answers show inside the agent's screen
  - [ ] Pass masked data only, and test the links end to end
  - [ ] Write down what happens if the server stops: chats and staff questions go back to today's tools until it is fixed
- **Output:** a working server with energy readings, and tested links to Globe's care and staff tools by month 7.

2.8 Staff trial. Lead: product owner, with care operations. Months 7 to 8.

- **Needs first:** the gateway from 2.6, the task-ready Core model and the interim safety sign-off from 2.5, the links from 2.7, about 500 staff and 30 to 50 care agents signed up, a short user guide, and the agreed trial targets and baselines.
- **Do:**
  - [ ] Give the staff assistant to about 500 Globe staff
  - [ ] Give suggested replies to 30 to 50 care agents, who check every one before it goes out
  - [ ] Collect usage, quality ratings, cost per answer and time saved every week, against the baselines
  - [ ] Fix the top issues
- **Output:** staff trial results, measured against the targets.

2.9 Privacy update and model card. Lead: privacy and AI governance lead. Month 8.

- **Needs first:** test results from 2.5, energy readings from 2.7, and the training records.
- **Do:**
  - [ ] Update the privacy impact assessment for training and the staff trial
  - [ ] Brief the National Privacy Commission on the plan, with legal
  - [ ] Hold an ethics board meeting on the results so far
  - [ ] Draft the model card: what it learned from, how it was tested, its limits, and its energy and carbon
- **Output:** an updated assessment and a draft model card.

2.10 Gate 2 pack. Lead: programme lead. Month 8.

- **Needs first:** the outputs of 2.4 to 2.9, and cybersecurity's review from 2.6.
- **Do:**
  - [ ] Run the locked golden sets on the release candidate
  - [ ] Bring the evidence together for each Gate 2 item
  - [ ] Confirm the go-live plan for Step 3
  - [ ] Hold the Gate 2 meeting
- **Output:** a go or stop decision for Step 3.

Milestones

Month	What is done
4	Two new people start; first data licences signed; server installed at STT GDC; gateway build starts
5	First corpus release; Filipino training of the Core model starts
6	Full corpus; private links to the care and staff test systems live
7	Task-ready Core model; interim safety sign-off; staff trial starts
8	Staff trial results; privacy update; Gate 2 meeting

What it needs and what it costs

Needs: about 30,000 hours of rented GPU time for Filipino training, about 10,000 more for tuning in the Philippines and for trials, one rented 8-GPU server in the locked space, licensed Filipino text, Filipino annotators, and time from Globe's care, legal, security and network teams.

Cost line	What it pays for	US$k
People	12 people for 5 months: 60 person-months at US$13,750	825
Training computing	40,000 GPU-hours at US$6.20: teach Filipino 30,000, first tuning 5,000, trials and reruns 5,000	248
Serving and outside AI	One 8-GPU server for 5 months at US$5.50 per GPU-hour (161), and outside AI as a rare backup (20)	181
Data	Licensed Filipino text from publishers and broadcasters (250), worked examples written by linguists (150)	400
Platform	Gateway and router (150), links to Globe's care and staff tools (100)	250
Governance	Regulator and legal work (50), ethics board (20)	70
Margin for surprises	15% of the lines above	296
Total		**2,270 (₱142M)**

What Globe has at the end

- A licensed Filipino training corpus and thousands of worked examples
- A Core model that speaks Filipino, Taglish and Cebuano and knows Globe's tasks
- A gateway and router on one server, linked to Globe's care and staff tools
- Results from a staff trial with about 500 staff and 30 to 50 care agents
- An updated privacy assessment and a draft model card

Gate 2: ready for customers? (end of month 8)

Every item must be true to release Step 3 (US$1.35M).

- [ ] The Core model meets its targets on the locked golden sets in Filipino, Taglish and Cebuano (evidence: locked golden set results in the Gate 2 pack; confirmed by the safety lead)
- [ ] The staff trial meets its targets for quality, safety, cost per answer and staff use (evidence: the trial report; confirmed by the product owner and care operations)
- [ ] The links to Globe's care and staff tools work end to end, with masked data only (evidence: the test report; confirmed by the platform engineer)
- [ ] Globe's cybersecurity team has passed the security review of the gateway, router, server and links (evidence: the review; confirmed by cybersecurity)
- [ ] Energy per 1,000 answers is within target (evidence: the metering report; confirmed by the MLOps and platform engineer)

If the answer is not yet: the team fixes the gaps and tests again within one to two months, and the Step 3 money stays locked until the gate is passed. If Globe stops here, it has spent about US$3.4M and keeps the model, the data and the platform.

8. Step 3: Go live (months 9 to 12)

Goal: fix what the staff trial found, pass outside safety and security checks, and open care chat to customers a few at a time, so Luntian is live at month 12. **Time:** 4 months. **Money:** US$1.35M (₱84M). **Team:** 12 people.

What must be in place first

- [ ] Gate 2 is passed and the Step 3 budget of US$1.35M is released
- [ ] An outside red-team firm and a penetration-test firm are contracted, with dates in months 10 and 11
- [ ] The go-live plan is agreed with care operations: which chat channel goes first, what share of traffic, and who checks answers
- [ ] A care workforce plan is agreed with HR and care operations: how agents' work changes, and the training they get
- [ ] Care agents and team leads are booked for training on how to work with Luntian, including how to report a bad answer
- [ ] Customer notices say when customers are talking to an AI, and how to reach a person
- [ ] A rollback plan is written: how care switches back to the old way within minutes, including when the server stops
- [ ] The incident steps in section 15 are agreed, with names on the call list

The work, activity by activity

3.1 Fix what the staff trial found. Lead: research and model lead. Months 9 to 10.

- **Needs first:** the staff trial results, the list of weak answers, and about 10,000 booked GPU-hours: 5,000 in the Philippines for tuning on the trial feedback, and 5,000 for trials.
- **Do:**
  - [ ] Add worked examples for the weak spots
  - [ ] Tune the Core model again on the staff trial feedback
  - [ ] Test on the working golden set, and keep the best version
- **Output:** a release candidate for customers.

3.2 Safety examples and safety filters. Lead: evaluation and safety lead. Months 9 to 11.

- **Needs first:** the red-team findings so far, Filipino and Taglish speakers to write safety examples, and time from the platform team.
- **Do:**
  - [ ] Write safety examples in Filipino, Taglish and Cebuano, and use them to tune and test
  - [ ] Add safety filters to the gateway that screen every question and answer in Filipino and Taglish
  - [ ] Test the filters against the list of harms agreed with the ethics board
- **Output:** safety filters in the gateway, and a larger safety test set.

3.3 Outside red team and penetration test. Lead: evaluation and safety lead, with Globe cybersecurity. Months 10 to 11.

- **Needs first:** the release candidate from 3.1, the safety filters from 3.2, and the contracted firms.
- **Do:**
  - [ ] Have an outside firm red-team the service in Filipino, Taglish and Cebuano
  - [ ] Run a penetration test on the gateway, the server and the links
  - [ ] Fix every critical and high finding, then test again
- **Output:** a red-team report and a penetration-test report, with no critical or high findings left open.

3.4 Go-live tests and monitoring. Lead: MLOps and platform engineer, with the linguists. Months 9 to 11.

- **Needs first:** the release candidate from 3.1, and the go-live plan agreed with care operations.
- **Do:**

  - [ ] Build the go-live test sets for care chat
  - [ ] Set up monitoring and alerts for quality, safety, cost and energy, with named people on call
  - [ ] Run a load test for the expected peak, and a rollback test
- **Output:** go-live tests passed, and monitoring in place.

3.5 Prepare care. Lead: product owner, with care operations and HR. Months 10 to 11.

- **Needs first:** the care workforce plan, the trial lessons, and the customer notices.
- **Do:**
  - [ ] Train agents and team leads
  - [ ] Update scripts and customer notices
  - [ ] Hold a final rehearsal
- **Output:** care ready for go-live.

3.6 Go live in care chat. Lead: product owner, with care operations. Months 11 to 12.

- **Needs first:** the go-live readiness checklist below, fully ticked.
- **Do:**
  - [ ] Month 11: go live on one care chat channel with a small share of traffic, with a person checking high-risk answers
  - [ ] Raise the share step by step while the targets hold
  - [ ] Review quality, safety and satisfaction every day for the first month, then every week
  - [ ] Month 12: Luntian handles its agreed share of care chat
- **Output:** Luntian live in customer care chat at month 12, in Filipino, Taglish and Cebuano.

3.7 Publish the model card. Lead: privacy and AI governance lead. Month 12.

- **Needs first:** the final test results, the red-team report and the energy readings.
- **Do:**
  - [ ] Publish the model card
  - [ ] Update the privacy impact assessment for customer use
- **Output:** a published model card and an updated assessment.

3.8 Plan the first year of running it. Lead: programme lead. Month 12.

- **Needs first:** real running costs since month 4, the go-live results, and the list of add-ons in section 9.
- **Do:**
  - [ ] Confirm the run team of 8 and the US$2.74M yearly budget (section 9)
  - [ ] Set the dates for the yearly refresh and the yearly checks

  - [ ] List which add-on triggers are close to being met
  - [ ] Prepare the decision on renting or buying the server (section 14)
- **Output:** an approved run plan and budget for months 13 to 24.

Go-live readiness (before customers see it)

All must be true before Luntian answers customers directly:

- [ ] Gate 2 is passed, and the release checklist in section 11 is ticked for this model version
- [ ] The load test shows the server handles the expected peak, with room to spare
- [ ] The rollback test passed: if the server stops or misbehaves, care switches back to today's system within minutes
- [ ] The outside red team and the penetration test left no critical or high findings open
- [ ] Agents are trained, and a person checks high-risk answers
- [ ] Customers are told when they are talking to an AI, and can reach a person
- [ ] Monitoring and alerts are on for quality, safety, cost and energy, with named people on call
- [ ] The incident steps in section 15 have been rehearsed once

Milestones

Month	What is done
9	Fixes from the staff trial start; safety examples written
10	Safety filters in place; outside red team starts; agents trained
11	Penetration test passed; go-live readiness checklist ticked; first customers on one chat channel
12	Luntian live in care chat; model card published; run plan approved

What it needs and what it costs

Needs: about 10,000 GPU-hours, the same 8-GPU server, outside red-team and penetration-test firms, Filipino writers for safety examples, and time from Globe's care, HR, security and legal teams.

Cost line	What it pays for	US$k
People	12 people for 4 months: 48 person-months at US$13,750	660
Training computing	10,000 GPU-hours at US$6.20: tuning on the staff trial feedback 5,000, trials 5,000	62
Serving and outside AI	The 8-GPU server for 4 months (128), and outside AI as a rare backup (20)	148
Data	Safety examples (50) and go-live test sets (40)	90
Platform	Safety filters (60) and monitoring (50)	110
Governance	Outside red-team audit (70) and penetration test (30)	100
Margin for surprises	15% of the lines above	176
Total		**1,346 (₱84M)**

What Globe has at the end

- Luntian live in customer care chat, in Filipino, Taglish and Cebuano
- A staff assistant in daily use
- Safety filters, an outside red-team report and a penetration-test report
- Monitoring, alerts and a tested rollback
- A published model card and an approved run plan

After month 12

There is no money gate at month 12. The running cost from month 13 is approved as part of Globe's normal yearly budget, and each add-on in section 9 needs its own yes.

9. After go-live: run and grow (from month 13)

From month 13, a run team of 8 keeps Luntian running and up to date for about US$2.74M (₱171M) a year. Everything else is an add-on: each one starts only when its trigger is met, its own numbers show that it pays, and the sponsor says yes.

Running the service

The run team

Role	People	What they do
Service owner	1	Runs Luntian as a Globe service: budget, targets, suppliers and reports
Model engineers	2	The yearly refresh, tuning and fixing weak spots
Data engineer	1	New data, masking and the data inventory
Filipino linguist	1	Worked examples and test questions
Platform engineers	2	The server, the gateway, monitoring, energy and cost
Safety and privacy lead	1	Tests, red teaming, privacy checks and audits

R.1 Daily checks. Lead: platform engineers, with the safety and privacy lead. Every day.

- **Do:**
  - [ ] Check quality, safety, cost and energy on the dashboards
  - [ ] Act on alerts, and roll back if an answer pattern looks wrong
  - [ ] Log bad answers reported by agents, and add them to the test sets
- **Output:** a daily check in the operations log.

R.2 The yearly refresh. Lead: model engineers. Once a year, or as two smaller updates.

- **Needs first:** a newer open model that beats the live one on Globe's tests, about 35,000 booked GPU-hours, and the lessons from live use.
- **Do:**
  - [ ] Update the data and test sets with lessons from live use
  - [ ] Repeat the eight training steps in section 11: about 25,000 GPU-hours of continued training on licensed and public text, and about 10,000 on GPUs in the Philippines for tuning on Globe data
  - [ ] Compare the new version with the live one, and release it only if it is better
- **Output:** a new version of the model each year.

R.3 Yearly checks. Lead: safety and privacy lead. Every year.

- **Do:**
  - [ ] Review the privacy impact assessment
  - [ ] Run the outside audit and a penetration test
  - [ ] Renew data licences that are due
  - [ ] Report to the ethics board
- **Output:** a yearly assurance pack for the sponsor.

R.4 Energy report. Lead: platform engineers, with the sustainability team. Month 15, then every year.

- **Needs first:** energy readings since go-live, the Philippine grid carbon factor, and renewable supply records from STT GDC.
- **Do:**
  - [ ] Work out energy and carbon for the model and per 1,000 answers, both location-based and market-based (section 16)
  - [ ] Get outside assurance
  - [ ] Add an AI energy line to Globe's Integrated Report
- **Output:** Luntian's energy use reported each year.

R.5 The server decision. Lead: service owner, with procurement and finance. Month 13.

- **Needs first:** the server's real use since go-live, and supplier quotes from 1.9.
- **Do:**
  - [ ] If the server is busy, compare renting (about US$0.39M a year) with buying (about US$0.6M plus space, power and support)
  - [ ] Buy only if buying is cheaper within about two years (section 14)
- **Output:** a rent-or-buy decision, signed by the sponsor.

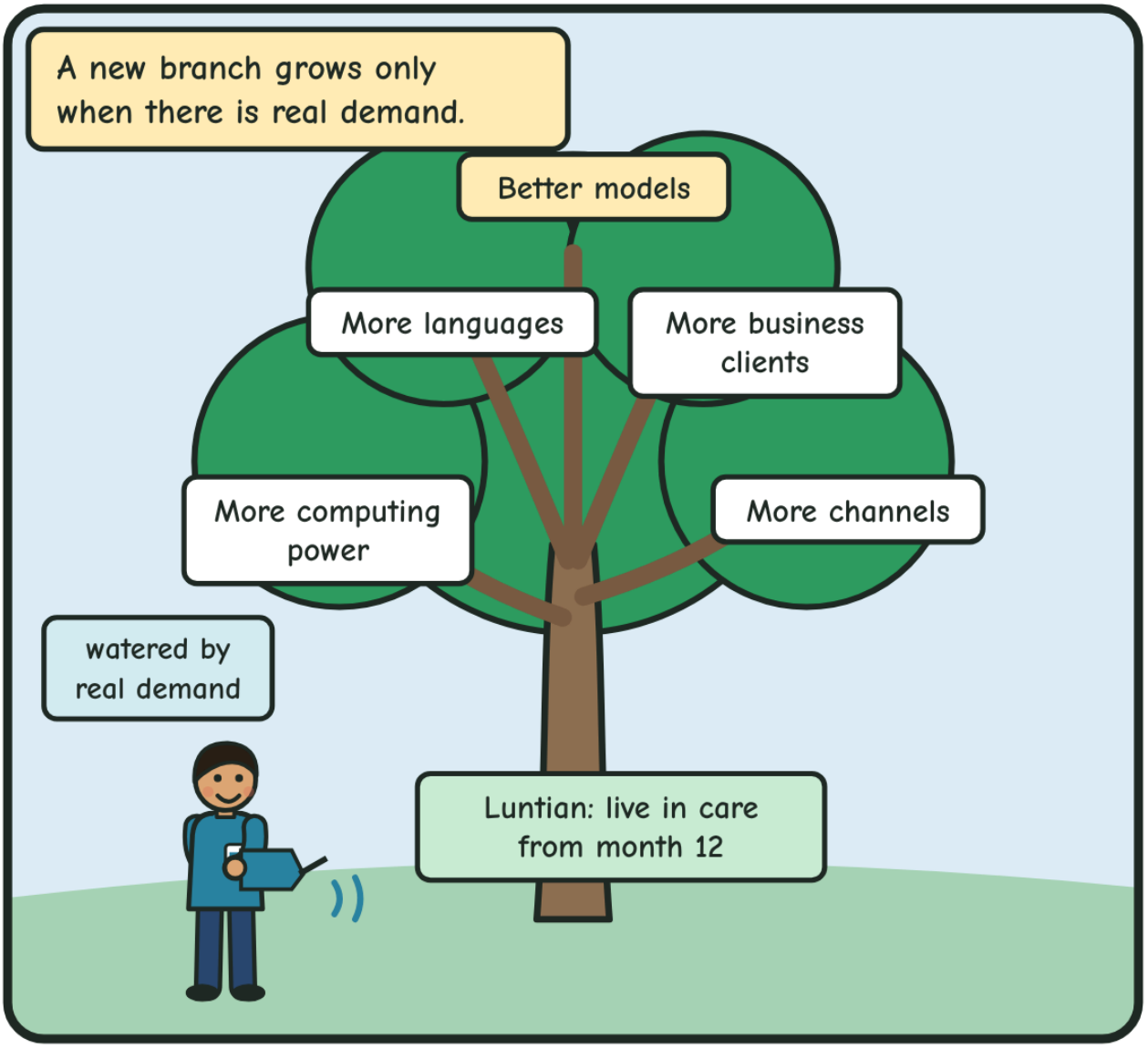
The yearly calendar

When	What happens	Who
Every day	Quality, safety, cost and energy checks	Platform engineers, safety and privacy lead
Every quarter	Ethics board meeting, report to the sponsor, and a rollback drill	Safety and privacy lead, service owner, platform engineers
Once a year	The model refresh, if a newer open model beats the live one, or two smaller updates	Model engineers
Every year	Privacy assessment review, outside audit, penetration test, licence renewals, and the energy line in the Integrated Report	Safety and privacy lead, platform engineers
Every two years	A review of the long-term path: keep adapting open models, build Globe's own from scratch, or join a national effort	Service owner, with the sponsor

What running it costs, about US$2.74M a year

Line	What it pays for	US$k a year
People	8 people all year: 96 person-months at US$13,750	1,320
Serving	One 8-GPU server all year: 8 GPUs × 8,760 hours × US$5.50, rounded up	386
Outside AI backup	Rare, very hard questions only, never with personal data	60
Model refresh	35,000 GPU-hours at US$6.20: one full refresh a year, or two smaller ones	217
Data	New licensed text and fresh test sets	150
Platform	Software support and monitoring	150
Governance	A yearly outside audit and the ethics board	100
Margin for surprises	15% of the lines above	357
Total		**2,740 (₱171M)**

Growing: the add-ons



The rule: an add-on starts only when its trigger is met, its own business case shows that the extra value covers the extra cost, and the sponsor says yes. The costs below are upper-side and include the 15% margin. Costs marked once do not include any running cost the add-on adds later.

Add-on	Starts only when	Extra cost
A1. A second server (8 more GPUs)	The first server is busy more than 70% of the time for four weeks	About US$0.44M a year
A2. Luntian for business clients	At least three banks, hospitals or agencies sign letters of intent. Each client's own servers are paid from its fees	About US$0.40M once

Add-on	Starts only when	Extra cost
A3. A standby site and ISO/IEC 42001 certification	Business clients or regulators ask for them	About US$0.28M once
A4. Voice on the care hotline	Care chat meets its targets for three months in a row	About US$0.80M once
A5. More languages: Ilocano, Hiligaynon and Waray	There is enough licensed data and the tests are good	About US$0.37M once
A6. Scam warnings in Filipino, from a small fast model	Globe's anti-scam team asks for it and the privacy review approves it	About US$0.26M once
A7. A large model for the hardest questions	More than 5% of questions need the outside AI backup	About US$0.32M once, plus a server

Before any add-on starts

- [ ] Its trigger is met, with the evidence written down
- [ ] Its business case shows that the extra value covers the extra cost, with a named owner
- [ ] The people, computing and data it needs are booked
- [ ] Privacy, legal and security have reviewed it where it touches new data or systems
- [ ] The sponsor has approved its budget

What each add-on involves

A1. A second server. Lead: platform engineers. About 2 months.

- **Needs first:** about 15 kW more in the same locked space, and supplier quotes.
- **Do:**
  - [ ] Rent or buy a second 8-GPU server, using the rule in section 14
  - [ ] Install it next to the first, and split the traffic between the two
  - [ ] Test failover: if one server stops, the other keeps the service running
- **Cost basis:** 8 GPUs × 8,760 hours × US$5.50, plus the 15% margin.
- **Output:** twice the capacity, and no single point of failure.

A2. Luntian for business clients. Lead: Globe Business, with the product owner. About 6 months.

- **Needs first:** at least three signed letters of intent, a named Globe Business lead, and legal's view on contract terms.
- **Do:**
  - [ ] Design a private Luntian offer for business clients, on shared or dedicated servers

  - [ ] Set the price list and service levels
  - [ ] Check the rules each client sector follows, such as the Bangko Sentral ng Pilipinas rules on outsourcing and technology risk for banks, and health data rules for hospitals
  - [ ] Prepare a contract template, and set up a private data space or dedicated servers for each client
  - [ ] Tune a version for each client on its own data, in the Philippines, paid from its fees
- **Cost basis:** two people for 6 months (165), client separation and admin tools (120), legal templates and a security review (65), plus the 15% margin.
- **Output:** the first paying business clients.

A3. A standby site and ISO/IEC 42001 certification. Lead: platform engineers and the safety and privacy lead. About 6 to 9 months.

- **Needs first:** space at a second STT GDC campus such as Cavite, private links to Globe, and an accredited certification body.
- **Do:**
  - [ ] Set up a small standby copy of the service: a few GPUs, the model and the settings
  - [ ] Test the switch from Fairview to the standby site and back, then repeat the test every quarter
  - [ ] Set up the AI management system that ISO/IEC 42001 describes, run an internal audit, and pass the two-stage outside audit
- **Cost basis:** the standby site (100) and ISO/IEC 42001 preparation and audit (140), plus the 15% margin. Yearly surveillance audits follow.
- **Output:** a working standby site and an ISO/IEC 42001 certificate.

A4. Voice on the care hotline. Lead: model engineers, with care operations. About 6 months.

- **Needs first:** consent forms or licences for speech, a recording and transcription supplier, access to the hotline's phone systems, and the privacy lead's approval.
- **Do:**
  - [ ] Record and transcribe Filipino, Taglish and Cebuano speech, with voices anonymised where needed
  - [ ] Build speech to text and text to speech around the Core model
  - [ ] Connect it to the hotline, test it with real users, and roll it out step by step
- **Cost basis:** speech data (220), 20,000 GPU-hours (124), two people for 6 months (165), the link to the hotline (120), and safety and quality tests (60), plus the 15% margin.
- **Output:** a voice bot on the care hotline.

A5. More languages. Lead: data engineer, with the linguists. About 4 to 6 months.

- **Needs first:** licensed text in Ilocano, Hiligaynon and Waray from universities and broadcasters, and linguists who speak them.
- **Do:**
  - [ ] Collect, clean and mask the text
  - [ ] Build a test set for each language
  - [ ] Tune the Core model, and release each language only when its tests pass
- **Cost basis:** data (150), 20,000 GPU-hours (124) and linguists (48), plus the 15% margin.
- **Output:** Luntian in three more Philippine languages.

A6. Scam warnings in Filipino. Lead: model engineers, with Globe's anti-scam team. About 6 months.

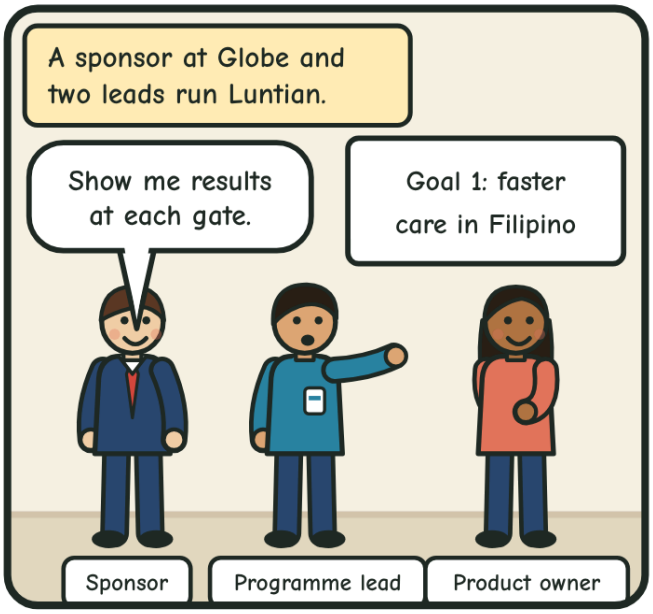
- **Needs first:** labelled scam-text examples from Globe's fraud team with personal details removed, and approval from legal and privacy of how the filter reads messages, what it keeps and for how long.
- **Do:**
  - [ ] Train the small Edge model to copy the Core model on scam checks (this is called distillation), and on the labelled examples
  - [ ] Run it inside Globe's network first, then test it in the GlobeOne app with the app team
  - [ ] Test it with real users before wider use
- **Cost basis:** 15,000 GPU-hours (93), one person for 6 months (83) and labelled examples (50), plus the 15% margin.
- **Output:** scam warnings in Filipino.

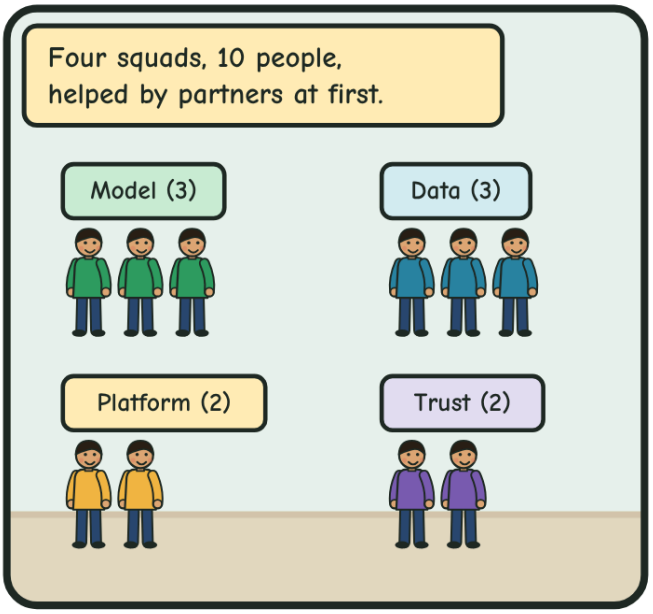
A7. A large model for the hardest questions. Lead: research and model lead. About 4 months.

- **Needs first:** evidence that more than 5% of questions go to the outside AI, a second server (A1), and about 45,000 booked GPU-hours.
- **Do:**
  - [ ] Adapt a 27B to 32B open model in the same way as the Core model
  - [ ] Send only the hardest questions to it, and use the outside AI less
- **Cost basis:** 45,000 GPU-hours at US$6.20 (279), plus the 15% margin, plus the second server.
- **Output:** fewer questions sent outside Globe, and better answers on hard cases.

10. The team, hiring and handover

Ten people start Luntian in Step 1, and two more join in Step 2, in four squads inside AIDE, Globe's in-house AI group, led by a programme lead and a product owner. After go-live, a run team of 8 keeps it going. Up to half of the roles can come from AIDE and Globe IT.





The core team

Role	People	What they are responsible for	Joins	Where they come from
Programme lead	1	The plan, budget, suppliers, gates and reports to the sponsor	Step 0	A Globe manager or a contracted lead
Product owner	1	Uses, KPIs, and adoption in care and with staff	Step 1	Globe staff
Research and model lead	1	Picks the base model and the training recipes	Step 1	New hire, helped by AI Singapore at first
Model-training engineers	2	Continued training, fine-tuning and safety tuning	Step 1	AIDE or new hires, in a joint team with AI Singapore
Data engineer	1	Data pipelines, masking personal data, and tracking where each record came from	Step 1	Globe IT or a new hire

Role	People	What they are responsible for	Joins	Where they come from
Filipino linguists	2	Cleaning text, writing worked examples and test questions	1 in Step 1, 1 in Step 2	New hires, with universities
MLOps and platform engineers	2	The server, the gateway and router, links to Globe systems, monitoring, and energy and cost tracking (GreenOps and FinOps)	1 in Step 1, 1 in Step 2	Globe IT or new hires
Evaluation and safety lead	1	Tests, red teaming and the safety filters	Step 1	New hire
Privacy and AI governance lead	1	Privacy assessments, contact with the privacy regulator, and the ethics board	Step 1	Globe's data privacy office

The squads: **Model** (the research lead and two engineers), **Data** (the data engineer and two linguists), **Platform** (the two MLOps and platform engineers) and **Trust** (the safety lead and the privacy lead). That is 10 people, plus the programme lead and the product owner.

The hiring plan

Roles	Open the roles	Start	Why then
Programme lead	Before Step 0	Step 0	Runs the planning study
The other 9 Step 1 roles	Named in Step 0, week 4, from AIDE, Globe IT, partners or contractors	Step 1, weeks 1 to 2	Outside hiring takes 8 to 12 weeks, longer than the time available
Permanent hires for Step 1 roles held by contractors	When Step 1 is approved, with offers that depend on Gate 1	Month 4, as Step 2 starts	They take over from contractors as Step 2 starts
The 2 Step 2 roles	Step 1, week 6, with offers that depend on Gate 1	Month 4	Plan on 8 to 12 weeks to hire specialists

Roles	Open the roles	Start	Why then
The run team of 8	Step 3, month 11	Month 13	Mostly people already on the team, moving into run roles

Where an outside hire cannot start in time, a partner or contractor covers the role until the hire joins.

Before any role is opened:

- [ ] The headcount is in an approved step budget, or the offer depends on passing the next gate
- [ ] A job description lists the skills needed, including Filipino and Cebuano for linguists and red teamers
- [ ] HR has agreed the salary band
- [ ] An interview panel is named, with a partner expert for technical roles
- [ ] A first-30-days plan is ready: access, tools, training and a buddy

Globe teams that support the core team

Team	What they do
Executive sponsor, a Globe chief officer	Owns the budget and chairs each gate
Steering committee	Reviews each gate, each model release and each add-on
Finance	Releases money at gates, and checks the business case and costs
Procurement	Runs supplier quotes and contracts
Cybersecurity	The secure zone, access rules, the security review and penetration tests
Network team	Private links between STT Fairview and Globe's systems
Legal	Licences, partner contracts and data agreements
Data protection officer	Privacy assessments, breach reporting and contact with the privacy regulator

Team	What they do
HR	Hiring, salary bands, training and the care workforce plan
Care operations	The staff trial and the rollout in customer care
Globe Business	Sells private Luntian to business clients, if add-on A2 starts
Sustainability team	Signs off energy and carbon reports
STT GDC Philippines	Space, power, cooling and on-site support

Partners and what they do (none approached yet)

Partner	What they do	When
AI Singapore, Singapore's national AI programme	Co-develops the Filipino adaptation and shares know-how from its SEA-LION models	Step 1 as advisers, then Steps 2 and 3 in a joint team
Philippine universities, such as UP Diliman, Ateneo de Manila and De La Salle	Linguists, test sets, research and new talent	All steps
Publishers and broadcasters	Licensed text, and speech if add-on A4 starts	From Step 2
STT GDC Philippines	Renewable-backed hosting for the server	From Step 2
ACEN and other electricity suppliers	Renewable power, moving toward hourly matching	From Step 2
NVIDIA and open-source projects	Hardware, software and support	All steps
DOST and DICT	Public data, government pilots, and the option of a national effort	After go-live

Every partner contract assigns jointly built work to Globe, so the know-how stays in-house.

Handover: how the skills move to Globe

Stage	Who leads the work	What Globe staff do
Steps 1 and 2	Partners and new specialists	Work in the same teams and learn by doing
Step 3	Globe staff	Lead, with partners reviewing and coaching
After go-live	Globe's run team	Run it alone, with partners on call for the yearly refresh

Each handover has a checklist: the documents, the access, the runbooks, and a few weeks where the Globe owner does the task while the partner watches.

How the team works together

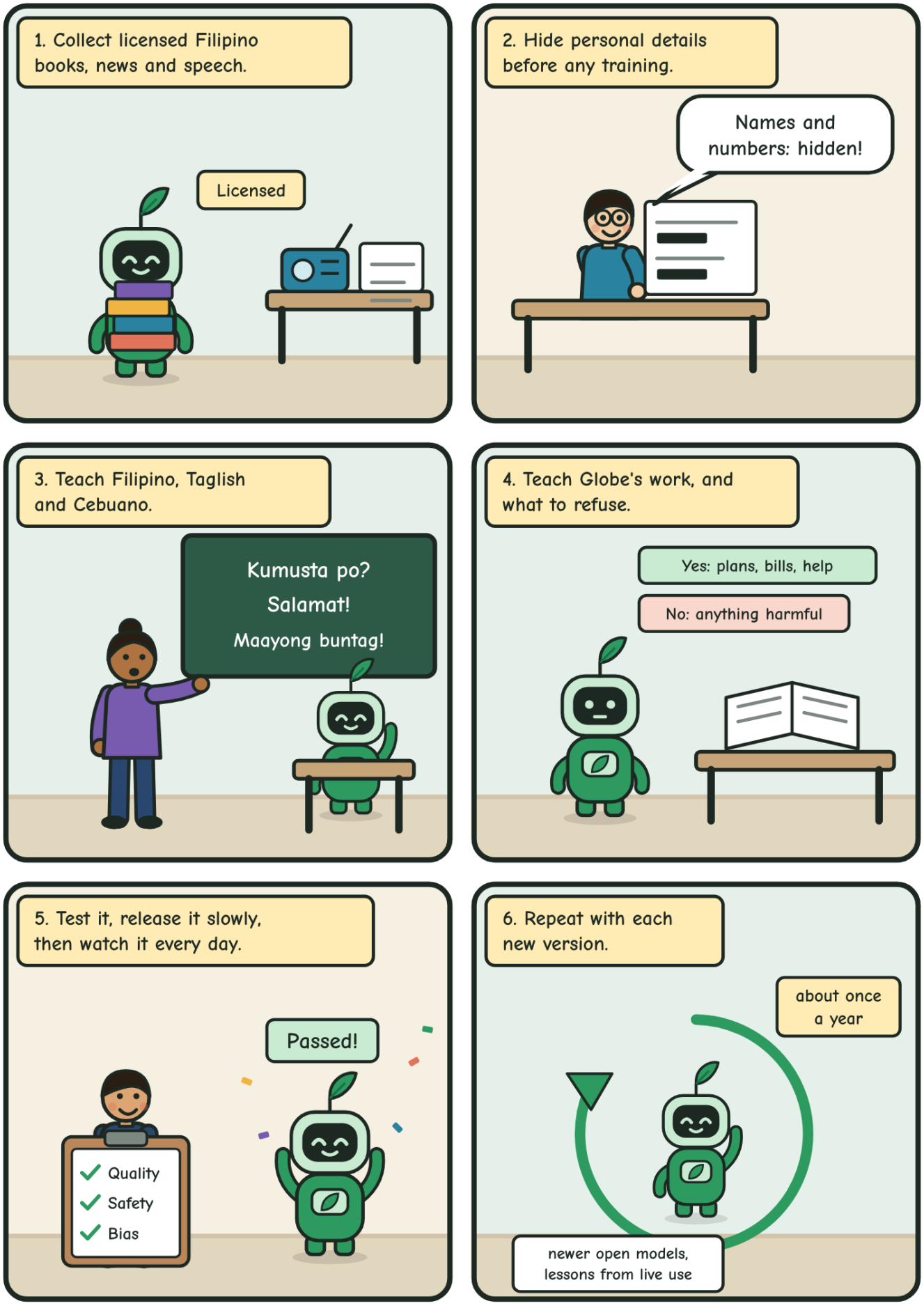
Meeting	How often	Who	What it decides
Squad stand-up	Daily, 15 minutes	Each squad	Today's work and blockers
Programme meeting	Weekly	Workstream owners and the programme lead	Progress, risks and cross-team issues
Steering committee	Every two weeks	The sponsor's delegate, programme lead, product owner, and the care, IT, privacy and finance leads	Priorities, spending and model releases
Sponsor update	Monthly	Executive sponsor and programme lead	Budget, KPIs and big risks
Ethics board	Every quarter, and before each gate	Board members, safety lead and privacy lead	Safety and fairness questions
Gate meeting	Months 3 and 8	Sponsor, Globe's leaders and the steering committee	Whether to release the next money
Add-on review	When a trigger is met	Sponsor and the steering committee	Whether to start an add-on

Who decides what

Decision	Who decides	Who prepares it	Who is consulted
Release money at a gate	Executive sponsor, with Globe's leaders	Programme lead	Steering committee, finance
Start an add-on	Executive sponsor	Service owner or programme lead	Steering committee, finance
Rent or buy the server	Executive sponsor	Platform engineers and procurement	Finance, cybersecurity
Use a new data source	Privacy lead	Data squad	Legal, ethics board
Release a new model version	Steering committee	Model squad	Trust squad, product owner
Safety sign-off	Evaluation and safety lead	Trust squad	Ethics board
Business prices and contracts	Head of Globe Business	Product owner	Finance, legal
Energy and carbon reports	Platform squad, signed off by Globe's sustainability team	Platform squad	STT GDC Philippines

11. Training the model

Every new version of Luntian goes through the same eight steps, so quality, privacy and energy are checked the same way each time. The first cycle runs in Steps 2 and 3, from month 5 to month 10, on about 50,000 GPU-hours. After go-live there is one full refresh a year, or two smaller ones, on about 35,000 GPU-hours a year, when a better open model is available.


1. Collect licensed Filipino books, news and speech.
Licensed
2. Hide personal details before any training.
Names and numbers: hidden!
3. Teach Filipino, Taglish and Cebuano.
Kumusta po?
Salamat!
Maayong buntag!
4. Teach Globe's work, and what to refuse.
Yes: plans, bills, help
No: anything harmful
5. Test it, release it slowly, then watch it every day.
Passed!
Quality
Safety
Bias
6. Repeat with each new version.
about once a year
newer open models, lessons from live use

Picking the starting model

In Step 1, five or six open models are tested side by side. The winner is the best mix of:

- **Quality** on Globe's golden test questions in Filipino, Taglish and Cebuano, and on public tests such as FilBench and SEA-HELM
- **Licence:** commercial use is allowed, and Globe can own what it builds on top (legal checks each one)
- **Cost and energy** to run, measured per 1,000 answers
- **Size:** a mixture-of-experts model of about 26B to 30B that uses only 3B to 4B for each word, from a family with smaller and larger members in case the Edge or Expert add-ons start later
- **Support:** an active company or community behind it, with regular new versions

Gate 1 confirms the choice and a back-up.

The eight steps

Step	What happens	Who	Where it runs	Time per cycle	Computing
1. Collect	Licensed news, books and broadcast transcripts and public Filipino web text, plus masked Globe care transcripts and knowledge-base articles with a lawful basis, kept separately in the Philippines	Data engineer	Globe's secure zone in the Philippines	4 to 6 weeks the first time, 2 weeks after	None
2. Clean	Remove duplicates, spam, machine-translated text and offensive content; balance the languages	Data engineer, linguists	Globe's secure zone	2 to 3 weeks	Small

Step	What happens	Who	Where it runs	Time per cycle	Computing
3. Protect privacy	Mask names, phone and account numbers and addresses; keep SIM registration records out; the privacy lead spot-checks samples	Data engineer, privacy lead	Globe's secure zone	Runs with step 2	Small
4. Teach Filipino	Continued training on several billion tokens of licensed and public Filipino, regional-language and English text	Model engineers	A rented GPU cluster, which may use renewable-backed GPUs elsewhere in the region, because it uses only licensed and public text, with no personal data	1 to 3 weeks	About 30,000 GPU-hours the first time, about 25,000 in a yearly refresh
5. Teach tasks	Fine-tuning on worked examples written by Filipino annotators, and on the masked Globe set	Model engineers, linguists	GPUs in the Philippines only, because it uses Globe data	1 to 2 weeks	About 5,000 GPU-hours in Step 2 and 5,000 in Step 3, shared with step 6
6. Make it safe	Safety tuning, plus safety filters that screen questions and answers in Filipino and Taglish	Evaluation and safety lead	GPUs in the Philippines	2 to 3 weeks	Shared with step 5
7. Test	Golden sets, FilBench, SEA-HELM, telco tests, red teaming, bias checks and energy per 1,000 answers	Evaluation and safety lead	GPUs in the Philippines	2 weeks	Small
8. Release and watch	Publish the model card, roll out gradually, and watch quality, safety, cost and energy every day	MLOps and platform engineers	The 8-GPU server at STT Fairview	Ongoing	The serving server

GPUs in the Philippines. Steps 5 to 7 use Globe data, so they run on GPUs in the Philippines: the 8-GPU server in Globe's locked space at STT GDC, or GPUs rented for a few weeks from a GPU cloud that hosts in the country. Tuning hours are priced at the training rate of US$6.20 an hour, inside the training line.

Trials. Steps 2 and 3 each set aside about 5,000 GPU-hours for experiments and reruns.

Refreshes. A refresh needs less computing than the first cycle, because the data, worked examples and tests already exist, and newer open models already know more Filipino. The yearly refresh is planned at about 35,000 GPU-hours: about 25,000 for continued training, and about 10,000 for tuning in the Philippines.

Before each training run

- [ ] The privacy lead has cleared every data source in the run, and the source list is recorded
- [ ] Each licence allows AI training, and the licence register is up to date
- [ ] The locked golden test sets are kept apart from all training data
- [ ] Any run that uses Globe data is booked on GPUs in the Philippines
- [ ] GPU time is booked and the run fits the step budget
- [ ] Experiment tracking and energy metering are switched on
- [ ] Base-model files are checked against their published fingerprints before use

Before each release: the release checklist

Every new model version passes all of these before any customer sees it:

- [ ] It meets or beats the targets on the locked golden sets in Filipino, Taglish and Cebuano
- [ ] It does at least as well as the live version on the same tests
- [ ] Safety tests pass, and no critical red-team issue is open
- [ ] Bias checks are done, and the results are written up
- [ ] Energy per 1,000 answers and cost per answer are measured and within target
- [ ] The model card is written: what it learned from, how it was tested, its limits, and its energy and carbon
- [ ] A rollback to the previous version is tested
- [ ] The safety lead has signed off, and then the steering committee has approved
- [ ] The rollout starts with a small share of traffic, with a person watching

Tools: open training tools such as NVIDIA NeMo or MosaicML Composer, vLLM or TensorRT-LLM for serving, and SEA-HELM and FilBench for testing. The Step 1 bake-off confirms the final choice.

12. The data Globe needs

Luntian needs five kinds of data at the start, and speech only if the voice add-on starts. Every record follows the same privacy rules before it is used. Licensed Filipino text matters most, because there is little good Filipino text on the open web.

Data	Examples	Rough target	Rights	Privacy handling
Licensed Filipino text	News archives, books and broadcast transcripts from Philippine publishers and broadcasters	Several billion tokens	A paid licence on the publisher's terms	Copyright terms logged; little personal data expected
Public text	Filipino web pages, government publications and open research collections	A few billion tokens (the Sailor2 team counted about 4.4B raw web tokens of Tagalog)	Public, with opt-outs respected	Filtered for personal data and quality
Globe data	Care chat and call transcripts, knowledge-base articles, network tickets	Up to a few billion tokens after filtering	A documented lawful basis and updated customer notices	Masked and kept to the minimum; used only for tuning, on GPUs in the Philippines; SIM registration data excluded
Worked examples	Answers and dialogues written by Filipino annotators, and safety examples	Thousands of examples at the start, growing each year	Made for Globe and owned by Globe	No real customer data
Test sets	Golden questions with agreed answers	About 1,000 at the start, growing each cycle	Owned by Globe	Kept apart from training data

Data	Examples	Rough target	Rights	Privacy handling
Speech, only with add-on A4	Recorded and transcribed Filipino, Taglish and Cebuano speech	Thousands of hours	Speaker consent or a licence	Voices anonymised where needed

The targets are planning figures; the data audit in Step 1 confirms them.

The privacy rules every record follows

1. A lawful basis is documented for every source, as the privacy regulator's 2024 advisory on AI expects.
2. A privacy impact assessment runs before first use, and again at each gate.
3. Fields that do not help the model, such as account numbers, are dropped.
4. Names, phone numbers and addresses are masked with neutral tags.
5. SIM registration records never enter training or search, as the SIM Registration Act requires.
6. Globe customer data is trained and served only in the Philippines.
7. Each record can be traced, so an objection or erasure request removes it from search at once and from the next training cycle.
8. The privacy lead spot-checks samples every cycle, and an outside audit runs every year.

Before any new data source is used

- [ ] The source has a named owner, inside Globe or at the supplier
- [ ] The lawful basis is written down, and the privacy impact assessment is updated
- [ ] For outside data, a signed licence allows AI training and lets Globe own the trained model
- [ ] Masking is tested on a sample, and the privacy lead has checked the result
- [ ] Each record carries a tag that shows where it came from
- [ ] The source is added to the data inventory and the licence register
- [ ] The privacy lead has approved it, after asking legal and the ethics board where needed

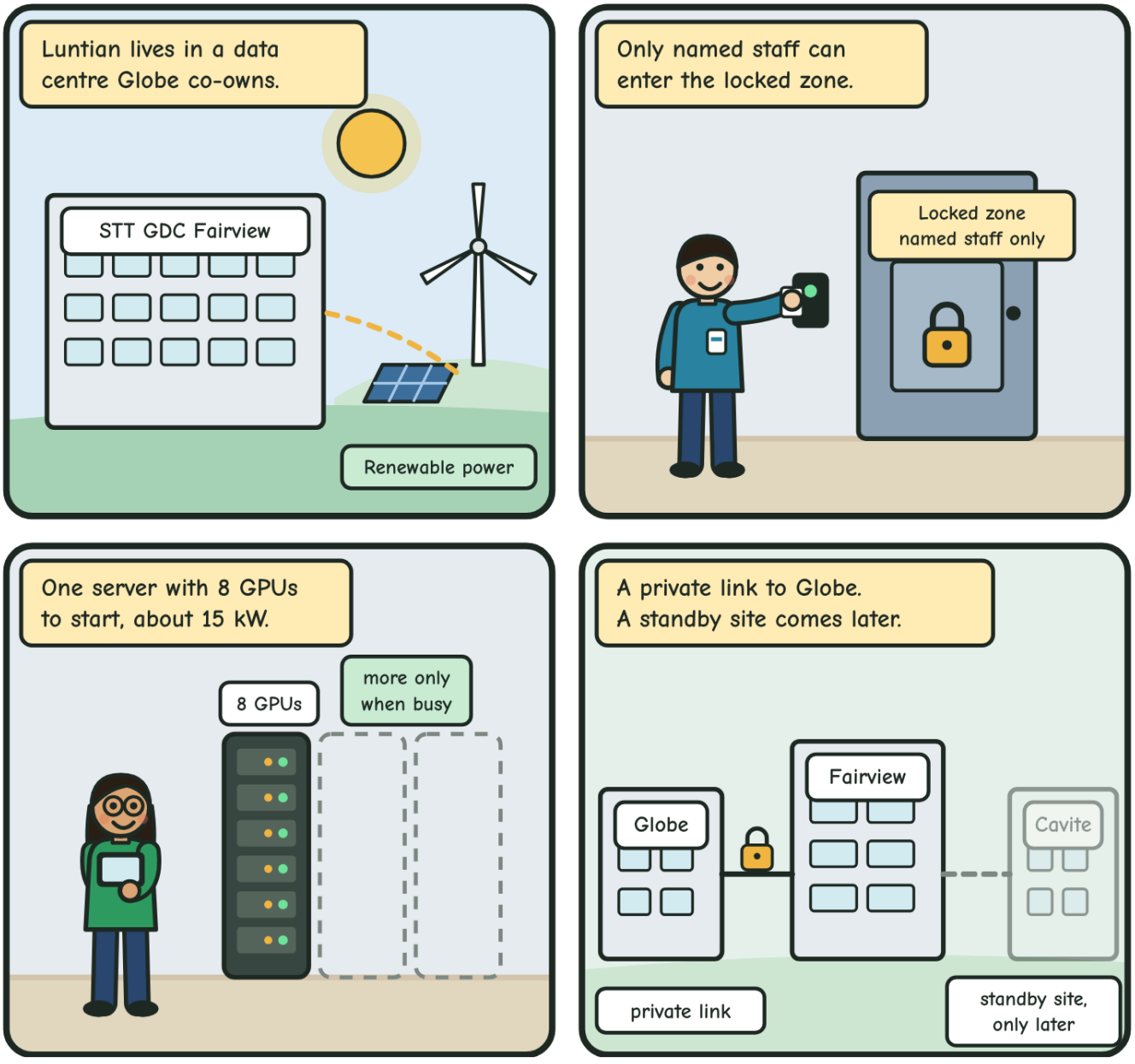
When someone objects or asks for their data to be erased

1. The request reaches the data protection officer through Globe's normal privacy channels.

2. The data team finds the person's records using the source tags.
3. The records come out of search indexes at once, and are marked so they leave the next training cycle.
4. The data protection officer replies to the person and records the case.

13. Setting up the data centre

Globe does not build a data centre. It rents one small locked space at STT GDC Philippines, the data centre company Globe co-owns, and puts one rented 8-GPU server there. More servers are added one at a time, only when the first is busy.



Where. STT Fairview 1 is partly live: its first floor of data halls is full, the second is being commissioned, and it supports liquid-cooled AI equipment on long-term renewable supply. STT GDC's Cavite campus is also live, and the company expects more than 30 MW

in operation by the end of 2026 (Globe, 7 August 2026).

How much space. One 8-GPU B200 server, such as NVIDIA's DGX B200, draws up to 14.3 kW, is air-cooled and takes 10 rack units (Flopper). The budget prices H100-class GPUs, whose 8-GPU servers draw about 10.2 kW (NVIDIA). With storage and network, Globe asks for about 15 kW, a quarter of the 60 kW in the earlier full plan, in a hall where more racks can be added next to it.

Who pays for the space. The serving rate of US$5.50 per GPU-hour includes hosting in the Philippines: the server, its locked space, power and cooling (section 17). Renting the space from STT GDC keeps that part of the spending inside a company Globe co-owns.

The eight set-up steps

#	Step	What happens	Needs first	When	Who
1	Agree space and power	About 15 kW of renewable-backed space at STT Fairview, with room for more racks later	Sizing from the bake-off, and STT GDC's indicative offer from Step 0	Step 1, weeks 4 to 10	MLOps and platform engineer, with STT GDC
2	Design the secure zone	A locked cage or private suite, a separate network, entry only for named people, cameras and visitor logs	Globe's security standards and the space offer	Step 1	Globe cybersecurity, with STT GDC
3	Rent the server	One 8-GPU server from a supplier that can host it in Globe's locked space	Three supplier quotes (section 14) and Gate 1	Ordered at Gate 1, installed in month 4	MLOps and platform engineer, with procurement
4	Connect to Globe	Private, encrypted links to Globe's care and staff systems, with no public internet path to the model	The secure zone design and the network team's capacity plan	Live by month 6	Globe network team
5	Install the software	Container platform, model serving, gateway and router, monitoring and energy meters	The server installed	Months 4 to 7	MLOps and platform engineers

#	Step	What happens	Needs first	When	Who
6	Test before go-live	A load test, a rollback test, a penetration test, and energy and PUE readings	Software installed, and test plans agreed with cybersecurity	Months 10 to 11	Platform engineers, cybersecurity
7	Go live	Live in care chat at month 12	The go-live readiness checklist in section 8	Month 12	Platform engineers
8	Grow only when busy	A second server in the same space (add-on A1), and a small standby site at a second STT GDC campus such as Cavite (add-on A3)	The trigger in section 9	When triggered	Platform engineers

If the server stops. With one server at the start there is no spare. A fault sends chats and staff questions back to today's care system and tools until the server is fixed or replaced, and this rollback is tested before go-live.

The secure zone checklist

- [ ] A locked cage or private suite, with entry only for named people
- [ ] Cameras, visitor logs, and escorts for suppliers on site
- [ ] A separate network, with no public internet path to the model
- [ ] Private, encrypted links to Globe's systems
- [ ] Multi-factor login for every person and system that reaches the server
- [ ] Encryption for stored data and for data on the network
- [ ] Audit logs kept and reviewed, with alerts on unusual access
- [ ] Energy meters on the rack, with readings sent to the energy and cost dashboard
- [ ] A written process for repairs, and for wiping disks before they leave the site

14. What to buy or rent, and how

Globe rents everything while it proves the idea. The one purchase decision is the server, at month 12, once Luntian is live and its use is known.

Item	What it is, simply	Rent or buy	When	Cost, upper side
Training computing	Powerful GPU chips rented by the hour for training: about 60,000 GPU-hours in the first year, then about 35,000 a year	Rent	Steps 1 to 3, then yearly	US$372k in the first year at US$6.20 an hour, then about US$217k a year
GPUs in the Philippines for tuning	Rented GPUs in the country, or the 8-GPU server, for the training steps that use Globe data	Rent	Steps 2 and 3, then yearly	Inside the training line, at US$6.20 an hour
Test server	A smaller GPU server in the Philippines for experiments and the two small pilots	Rent	Step 1	US$60k
The serving server	One 8-GPU server in Globe's locked space that answers staff and customers	Rent, then buy at month 12 if it is busy	Installed in month 4	About US$32k a month, or about US$0.39M a year, at US$5.50 per GPU-hour. Buying: about US$0.6M, plus space, power and support
Storage and network	Fast storage for data and model files, and the links inside the rack	Comes with the rented server	With the server	Inside the serving rate
Space, power and cooling	A locked space of about 15 kW at STT GDC Philippines' Fairview campus	Rent (colocation)	Agreed in Step 1	Inside the serving rate

Item	What it is, simply	Rent or buy	When	Cost, upper side
Software	Mostly open-source tools: NVIDIA NeMo for training, vLLM for serving, plus a model registry, search index, gateway and monitoring	Free, with optional paid support	Step 1	Inside the platform lines (US$560k to go live)
Filipino data	Licensed news, books and broadcast text	Licence	Step 2	US$250k, inside the data line
Annotators	Filipino speakers who write worked examples, safety examples and test questions	Contract	Steps 1 to 3	Inside the data lines (US$636k to go live)
Outside AI service	A contracted frontier model for rare, very hard questions, sent only masked requests with no personal data	Pay per use	From Step 1	US$40k in Step 1, US$20k in each of Steps 2 and 3, then US$60k a year
Audits	An outside red-team audit and a penetration test before go-live	Buy as services	Step 3	US$100k, inside the governance line
Add-ons	A second server, a standby site, ISO/IEC 42001, speech data and more	Only when triggered	After go-live	Section 9

Prices: B200 GPUs rented for US$6.25 to US$6.89 an hour on demand in October 2026, and complete 8-GPU B200 servers cost more than US$500,000 (Morph, October 2026). At those rates the first year's training would cost up to about US$413k, which fits inside the

15% margin. B200s cost more per hour than the H100-class GPUs in the budget, but in ML.ENERGY's 2026 tests they used a median 35% less energy per token (ML.ENERGY). The Step 1 capacity plan picks the type by cost per answer, not cost per hour.

Rent or buy, in plain words

- **Rent while Globe is still learning.** Needs change fast, and nothing sits idle.
- **Buy when steady use is expected.** Renting the 8-GPU server all year costs about US$0.39M. Buying one costs about US$0.6M once, plus space, power and support. If those stay under about US$0.1M a year, buying is cheaper after about two years.
- **When to decide:** at month 12 (activity R.5 in section 9), once Luntian is live and the server's use is known. Until then, nothing is bought.

How to buy, step by step

1. Write down the needs: number of GPUs, memory, power, delivery date, support and energy reporting.
2. Ask at least three suppliers to quote, such as server makers, NVIDIA partners and GPU clouds that can host in the Philippines.
3. Compare the full cost over the contract, the delivery date, energy use and support, not just the price.
4. Sign with service levels: uptime, repair times, and deletion of Globe data when the contract ends.
5. Run the staff trial workload as an acceptance test before the final payment.

Who runs it: Globe procurement, with the programme lead and the platform lead. Globe's cybersecurity team reviews the security terms, and the executive sponsor signs.

Start early. New suppliers may need to be set up in Globe's supplier system, so start at Gate 1 and the server can be installed in month 4. If Globe later decides to buy the server, capital approval may be needed, so start that in month 10.

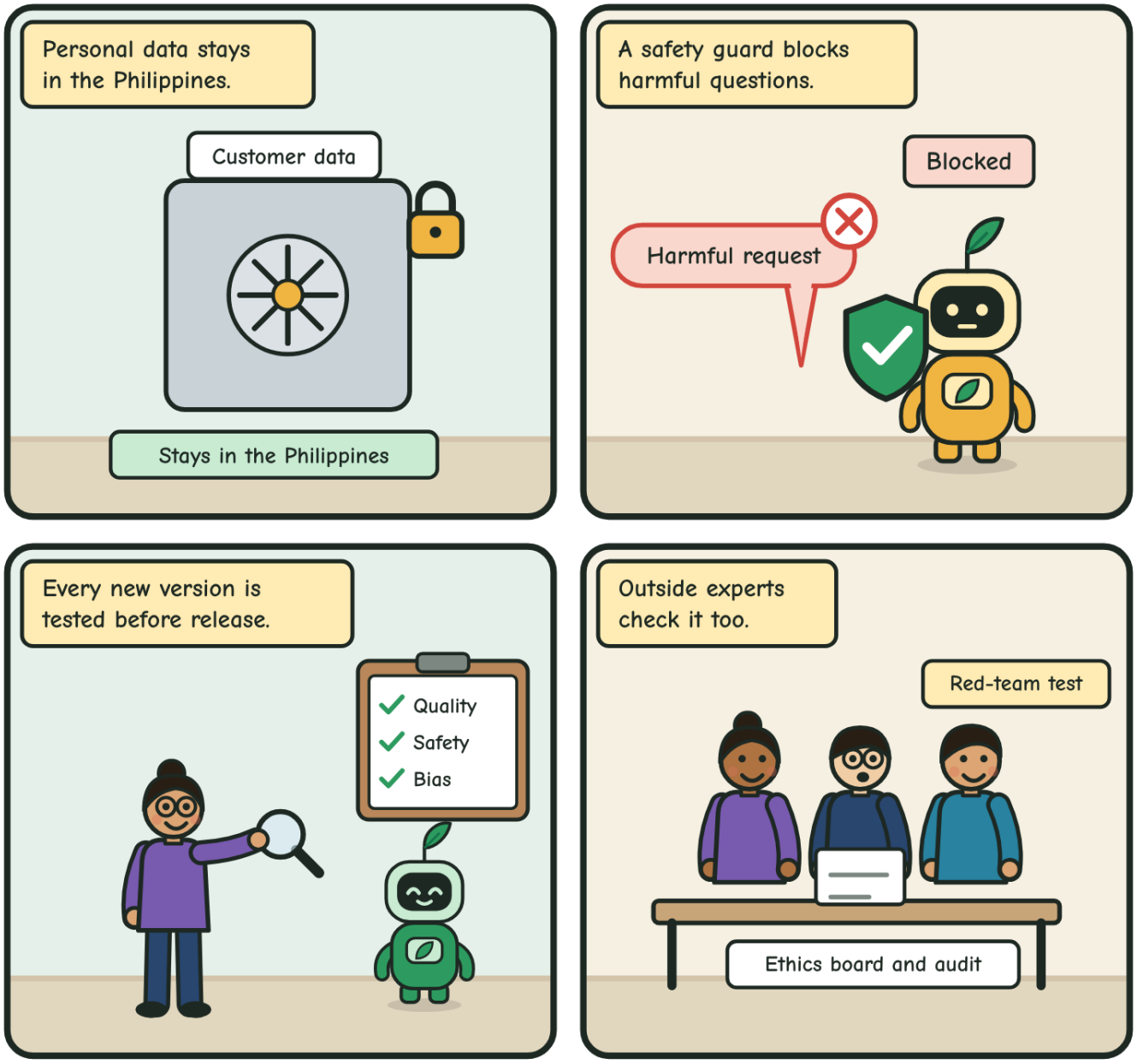
Before signing any supplier contract

- [ ] Anything built for Globe, or with Globe, belongs to Globe
- [ ] Personal data stays in the Philippines, and the supplier may not use any Globe data to train its own models
- [ ] Globe data is deleted, with proof, when the contract ends
- [ ] Service levels cover uptime, response times and repair times, with credits if they are missed
- [ ] Globe's cybersecurity team has reviewed the security terms, including who can reach what

- [ ] Globe may audit the supplier, or see independent audit reports
- [ ] The supplier reports energy use, for computing and hosting
- [ ] Exit terms let Globe move to another supplier without losing data or models
- [ ] Prices are fixed or capped for the contract term
- [ ] For the rented server: Globe can end the rental after a gate if the answer is no
- [ ] For the outside AI service: only masked requests with no personal data are sent, the supplier keeps nothing beyond what the contract allows, and the privacy lead has reviewed the transfer

15. Safety, privacy and security

Safe and in-house means four things: Globe owns what is built, personal data never leaves the Philippines, every model is tested before release, and the system is locked down like any critical Globe system.



What in-house means

- **Globe owns** the trained model (within the base model's licence), the Filipino datasets and test sets, the platform, and any work built with partners, which the contracts assign to Globe.
- **It runs** in Globe's secure zone at STT GDC Philippines, reachable only by named people.
- **Personal data** is trained and served only in the Philippines. The outside AI service gets only masked requests with no personal data.
- **The know-how stays.** Partners work in joint teams and hand over, and Globe staff run Luntian on their own from go-live.

The rules that never change

- **Personal data stays in the Philippines.** Anything that uses Globe's customer data is trained and run only in the Philippines.
- **Personal details are hidden first.** Names, numbers and addresses are removed before training, and SIM registration records are never used.
- **Only named people get in.** The server sits in a locked, private area, and only named staff can reach it or the data.
- **Nothing goes live untested.** No model answers customers directly until it passes the release checklist in section 11. In the pilots and the staff trial, an agent reviews and sends every reply to a customer.
- **People check risky answers.** High-risk answers are checked by a person, and every reply is logged.
- **The outside AI sees nothing private.** The outside service, used only for rare, very hard cases, never receives personal data.
- **Privacy is checked again at every gate**, following the Philippine privacy regulator's rules for AI.
- **Energy and carbon are measured and published** for every model.

The safety controls

Risk	What could go wrong	What Luntian does	Owner
Customer data leaks	Personal data shows up in answers or leaves the country	Masking before training; SIM registration records never used; search returns only what that user may see; encryption in storage and on the network; outside AI only for masked requests	Privacy lead, data engineer
Tricked by a prompt	Someone makes the AI ignore its rules or reveal its instructions	Safety filters screen questions and answers; the instructions hold no secrets; regular red teaming	Safety lead

Risk	What could go wrong	What Luntian does	Owner
Too much power	The AI takes an action it should not	Luntian answers and suggests; any action, such as changing a customer's plan, needs a check by a system or a person	Product owner, platform engineers
Wrong or harmful answers	False information or bias	Golden test sets, bias tests, gradual rollout, daily checks and quick rollback	Safety lead
Poisoned data or models	Bad data, or a tampered base model or tool	Licensed, checked sources; model files verified before use; supply-chain checks on software	Research lead, platform engineers
Break-ins	Hackers reach the server	Secure zone, separate network, multi-factor login, audit logs and yearly penetration tests	Globe cybersecurity, platform engineers
Runaway use	A flood of requests drives up cost or slows the service	Rate limits and spending caps for each channel	Platform engineers
Outage	The service goes down	With one server at the start, a fault sends chats back to today's care system; the rollback is tested before go-live; a second server or a standby site comes with add-ons A1 and A3	Platform engineers

These risks follow the OWASP Top 10 for LLM applications (2025), the checklist security teams use for AI systems (summary).

Outside checks

- An ethics board with outside members, set up in Step 1

- Privacy impact assessments at every gate, following the privacy regulator's AI advisory
- An outside red-team audit and a penetration test before go-live, then a penetration test and an outside audit every year
- ISO/IEC 42001 certification for managing AI when business clients or regulators ask for it (add-on A3), with a surveillance audit every year after that
- Published model cards and energy figures for every release

If something goes wrong

1. **Spot it.** A monitoring alert, a complaint, or a report from an agent or a staff member. Anyone can raise it.
2. **Contain it within minutes.** Switch off the affected feature, roll back to the last good version, or send the traffic to people. The person on call decides, and needs no approval to switch something off.
3. **Assess it.** The safety lead, the privacy lead and cybersecurity decide how serious it is, and whether personal data is involved.
4. **Report it.** If it is a personal data breach that must be reported, the data protection officer tells the National Privacy Commission and the affected people within 72 hours of learning of it. That deadline comes from the Commission's breach rules, Circular 16-03 (guide). The sponsor is told the same day.
5. **Fix and test.** Find the cause, fix it, add the case to the test sets and the red-team list, and pass the release checklist again before switching back on.
6. **Learn.** Write a short review within two weeks: what happened, why, and what changes. Share it with the steering committee and the ethics board.

16. Keeping it green

Luntian stays low-carbon through seven rules, and Globe measures and publishes the result for every model.

The scale. Training in the first year, about 60,000 GPU-hours, comes to about 67 tCO2e on the Philippine grid average. That is about 0.03% of Globe's 2025 Scope 1 and 2 emissions of 229,913 tCO2e (Globe). The yearly refresh adds about 39 tCO2e. The one 8-GPU server adds up to about 79 tCO2e a year at full load, and on contracted renewable supply the market-based figure is near zero. These figures assume H100-class GPUs at about 1.91 kWh per GPU-hour, including cooling, and 0.588 kgCO2e per kWh.

The seven rules

1. **Adapt, do not build.** Recalculated on the Philippine grid from published GPU-hours, training Llama 3.1 8B from scratch (about 1.46M GPU-hours) would emit about 1,640 tCO2e. Adapting a 9B model, as AI Singapore did for Gemma-SEA-LION-v3-9B (about 15,400 GPU-hours), emits about 17 tCO2e, roughly 99% less per model.
2. **Clean power only.** Train and serve on renewable-backed power. STT GDC Philippines already matches all its power with renewable supply or certificates (STT GDC).
3. **Small first.** One mixture-of-experts Core model does most of the work. In ML.ENERGY's 2026 tests, a mixture-of-experts model used about 3.6 times less energy per token than a standard model of similar size (ML.ENERGY).
4. **Think hard only when needed.** Reasoning modes use about 30 times more energy on average (AI Energy Score v2), so they stay off unless the router asks for them.
5. **Serve efficiently.** Grouping requests together (batching) cuts energy per token several times over.
6. **Cool efficiently.** A hall at STT Fairview that supports liquid cooling, with a target PUE of 1.35 or better.
7. **Measure and publish.** Energy and carbon per model and per 1,000 answers, using the SCI for AI and ITU-T L.1801 methods.

What Globe measures

Measure	How	How often	Where it is reported
Energy per training run	GPU and facility meters	Every run	The model card
Energy per 1,000 answers	Server meters divided by answers served	Daily	Operations dashboard and quarterly report
Carbon, location-based	Energy times the Philippine grid factor of 0.588 kgCO2e per kWh	Monthly	Integrated Report
Carbon, market-based	Energy times the carbon of the contracted renewable supply	Monthly	Integrated Report
PUE	Facility energy divided by computing energy	Monthly	STT GDC report

17. The budget

Going live costs US$4.79M (₱299M) in three steps, on upper-side estimates with a 15% margin on every line. After that, running Luntian costs about US$2.74M (₱171M) a year. The first two years, with no add-ons, come to about US$7.53M (₱471M), about half of the earlier US$14.41M full plan. People are the biggest cost, about 40% of the money to go live; computing is about a sixth.

Every cost line by step, US$k

Cost line	Step 1	Step 2	Step 3	To go live	Each year after
People	413	825	660	1,898	1,320
Training computing	62	248	62	372	217
Serving and outside AI	100	181	148	429	446
Data	146	400	90	636	150
Platform	200	250	110	560	150
Governance	100	70	100	270	100
Subtotal	1,021	1,974	1,170	4,165	2,383
Margin for surprises, 15%	153	296	176	625	357
Total	**1,174**	**2,270**	**1,346**	**4,790**	**2,740**
Total in pesos	**₱73M**	**₱142M**	**₱84M**	**₱299M**	**₱171M**

The line-by-line detail for each step is in sections 6 to 9. Figures are rounded to the nearest US$1,000.

How the money is released

Point	Money released	The most Globe can have spent
Step 1 approval	US$1.17M	US$1.17M at Gate 1
Gate 1, month 3	US$2.27M	About US$3.44M at Gate 2
Gate 2, month 8	US$1.35M	US$4.79M at go-live, month 12
From month 13	About US$2.74M a year, in Globe's normal yearly budget	Each add-on needs its own yes

Unit costs used

Item	Rate	Basis
One person for one month	US$13,750	US$165k a year, fully loaded: pay, benefits, hiring and partner experts. It is an upper-side blend of Manila and regional rates
Training, one GPU for one hour	US$6.20	CoreWeave's on-demand H100 list price of about US$6.16, rounded up. Newer B200 GPUs were listed at up to US$6.89 an hour in October 2026, which the 15% margin covers
Serving, one GPU for one hour	US$5.50	Nebius's H100 list price of US$4.50, plus 20% for hosting in the Philippines, rounded up. It covers the server, its locked space, power and cooling, and is a planning rate until suppliers quote
Exchange rate	₱62.5 per US$	The rate TelecomTV used for Globe's AI results in September 2026; check it against the Bangko Sentral ng Pilipinas rate when the budget is approved
Margin for surprises	15%	Applied to every line; if it is not needed, it is not spent

The add-ons. Each add-on in section 9 has its own cost and needs its own yes: a second server about US$0.44M a year; business clients about US$0.40M once; a standby site and ISO/IEC 42001 about US$0.28M once; voice about US$0.80M once; more languages about US$0.37M once; scam warnings about US$0.26M once; and a large model about US$0.32M once, plus a server.

When the money comes back

Value starts when Luntian goes live at month 12. Care and staff value builds up over six months, and business clients, if add-on A2 starts, join over the year from month 19. These values are planning assumptions, not yet checked against Globe's own data; activity 1.10 in Step 1 checks them.

Where the value comes from	How it works	Value a year, once running
Customer care	Luntian answers routine chats in the customer's own language and helps agents with the rest, with quality checked on every chat	About US$3.0M (₱188M)
Staff assistant	Globe staff use Luntian for sensitive work instead of paid outside AI tools	About US$1.6M (₱100M)
Business clients, later	Globe Business sells a private Luntian, at about US$500,000 of revenue per client a year and a 50% margin	About US$0.25M (₱16M) per client

What Luntian is used for	Value a year, once running	Pays back
Care and staff only	About ₱288M (US$4.6M)	Month 50: 38 months after going live, or just over 4 years after approval
Plus 10 business clients	About ₱444M (US$7.1M)	Month 39: 27 months after going live
Plus 20 business clients	About ₱600M (US$9.6M)	Month 33: 21 months after going live

The two business cases also pay for the business set-up (US$0.40M), the standby site and ISO/IEC 42001 (US$0.28M), and a second shared server from month 19 (US$0.44M a year). To pay back within two years of going live, Luntian must bring about ₱360M of value a year, reached in full by month 18, with no add-ons to pay for. The earlier full plan could not pay back on care and staff savings alone, because its running cost of US$4.8M a year was higher than those savings.

Two ways to spend less

- **Use Globe's own people.** If about half the team comes from AIDE and Globe IT, new spending to go live falls from about US$4.8M to about US$3.7M, and new spending on Step 1 to about US$0.94M, because their time is already paid for.
- **Buy the server once it is busy.** At month 12, buying the server instead of renting it is cheaper after about two years, if space, power and support stay under about US$0.1M a year (section 14).

Two more ideas to test in Step 1: paying some content owners through revenue shares instead of upfront licence fees, and using Singtel group GPU capacity for short training runs if its prices beat the market.

18. The numbers we track

Each gate is a check against numbers agreed in advance. In weeks 1 and 2 of Step 1, the sponsor and the product owner set a target for each measure below, and the targets go into the programme charter.

Measure	What it shows	How it is measured	First measured	Used at
Golden-set score	How good the answers are in Filipino, Taglish and Cebuano	Share of the locked test questions answered well, scored the agreed way	Step 1 bake-off	Gate 1 (Filipino and Taglish), Gate 2 (all three languages), every release
Open critical safety issues	Whether the AI is safe to release	Critical red-team findings not yet fixed; the target is zero	Step 2	Gate 2, go-live, every release
Staff use	Whether staff find it useful	Weekly active users of the staff assistant, and hours saved	Step 1 staff pilot	Gate 2 (staff trial targets), after go-live
Suggested-reply acceptance	Whether agents trust the answers	Share of suggested replies that agents send with no or small edits	Step 1 care pilot	Gate 2 (staff trial targets)
Cost per answer	Whether it costs less	Total running cost divided by answers served	Step 1 pilots	Gate 2, after go-live
Energy per 1,000 answers	Whether it is green	Server meters divided by answers served	Step 1 bake-off	Gate 2, after go-live

Measure	What it shows	How it is measured	First measured	Used at
Handling time	Time saved for agents	Average minutes per contact, against the baseline	Step 1 baseline	Gate 2 (staff trial), after go-live
Containment rate	How many care chats Luntian solves without a person	Share of chats closed without moving to an agent, with quality checked	Go-live, month 11	Each rise in the share of traffic, and the care savings after go-live
Customer satisfaction	Whether customers are happy with the answers	Globe's usual survey after a chat	Step 1 baseline	Each rise in the share of traffic, and after go-live
Uptime and response time	Whether the service is reliable	Share of time the service is up, and time to the first answer	Go-live	Every month after go-live
Privacy requests handled	Whether privacy promises are kept	Objections and erasure requests closed on time	Step 2	Gate 2, and every year after go-live
Add-on triggers	Whether an add-on should start	Server busy time, the share of questions sent to the outside AI, chat targets met, and signed letters of intent	After go-live	The triggers in section 9

Every target is a number, such as a minimum test score or a maximum cost per contact, so each gate compares results with agreed numbers.

19. Risks: what could go wrong, and what we do

The two biggest risks are too little Filipino data and a quality gap to the best global AI. Both have a plan, and every risk has an owner and an early warning sign.

Risk	Early warning sign	What we do	Owner
Not enough Filipino data	Licences not signed by month 5, or Cebuano scores lag the other languages	License publisher and broadcaster archives, write worked examples, and test each language separately	Data squad
Quality below the best global AI	Golden-set scores stall below target during training	Release only when the targets are met; send hard cases with no personal data to an outside AI	Trust squad
Newer open models arrive fast	A new open model beats Luntian on Globe's tests	Keep the data and tests reusable, and refresh once a year	Model squad
A privacy mistake	Masking misses found in spot checks, or a complaint	Privacy checks, masking, no SIM data, early talks with the privacy regulator, a yearly outside audit, and the incident steps in section 15	Privacy lead
Harmful or wrong answers in Filipino	Red-team findings rise, or agents flag bad answers	Safety filters, red teaming, and people checking high-risk answers	Trust squad
AI experts are hard to hire	No accepted offers for the two Step 2 roles by week 12 of Step 1	Joint teams with AI Singapore and universities, then handover; train AIDE staff	Programme lead
The one server fails	Alerts, or slow answers	Roll back to today's care system within minutes, tested before go-live; add a second server (A1) or a standby site (A3) when triggered	Platform squad

Risk	Early warning sign	What we do	Owner
GPU shortages or price rises	Supplier quotes above plan	Book the computing at Gate 1, keep the model small, and use the 15% margin	Platform squad
Value lower than planned	The Step 1 business case shows less value than the planning assumptions	Stop at Gate 1 having spent US$1.17M, or narrow the uses before Step 2	Product owner
Claims of greenwashing	Questions about how carbon is counted	Report both location-based and market-based carbon, get outside assurance, and plan for hourly matching	Platform squad, with the sustainability team
Low adoption	Staff trial usage falls after the first weeks	An owner and a KPI for each use, and the staff trial before the customer launch	Product owner

20. Words you may not know

Every technical term and name used in this playbook, each in one line.

Terms

Term	In plain words
AI model (LLM)	The "brain" behind tools like ChatGPT, which reads and writes text
Annotator	A person who writes or checks examples and test answers for training
B200 and H100	Two generations of NVIDIA's data centre GPUs; the B200 is newer, faster and needs more power per chip
Bake-off	A side-by-side test of candidate models on the same questions, to pick the best starting point

Term	In plain words
Base model	The open model Globe starts from, before adding Filipino and Globe knowledge
Batching	Grouping many requests together so the GPUs work more efficiently
BPO	Business process outsourcing: firms that run services such as call centres for other companies
Certification body	An accredited firm that audits a company against a standard such as ISO/IEC 42001 and issues the certificate
Colocation	Renting space, power and cooling in someone else's data centre for your own servers
Container platform	Software that packages programs so they run the same way on any server
Continued training	Training an existing model further on new text, such as Filipino, before teaching it tasks
Corpus	A large, organised collection of text used for training
Data centre	A secure building full of servers, with power and cooling
Data protection officer	The person at Globe responsible for privacy under the Data Privacy Act
DGX	NVIDIA's brand of complete AI servers, such as the DGX B200 with eight GPUs
Distillation	Training a small model to copy a larger one, so it answers almost as well at lower cost
Document search	Looking up Globe documents at answer time, so replies use current facts
Experiment tracking	A tool that records every training run: the data, the settings and the results
Failover	Switching to a back-up automatically when something breaks
Fine-tuning	Short, targeted training that teaches a model specific tasks or Globe's house style

Term	In plain words
Frontier model	One of the largest and most capable AI models, run by a few global companies
Gate	A checkpoint where leaders decide whether to release the next amount of money
Gateway	The front door that hides personal details and checks every request before it reaches a model
Golden set	A fixed set of real Globe questions with agreed answers, used to test every release; its locked part is used only at gates
GPU	A powerful computer chip used to train and run AI
GPU-hour	One GPU running for one hour, the basic unit of AI computing
GreenOps and FinOps	Tracking and cutting the energy and carbon (GreenOps) and the money (FinOps) that computing uses
Hourly matching	Buying clean power that matches use hour by hour, not just over a year
ISO/IEC 42001	An international standard for managing AI responsibly; companies can be certified against it
KPI	A number that shows whether something is working
Letter of intent	A signed, non-binding note from a client saying it plans to buy
Load test	A test that sends a heavy, realistic flow of requests to check that the system copes
Location-based and market-based	Two ways to count electricity emissions: at the local grid's average, or after contracted renewable supply
Masking	Replacing names, numbers and addresses with neutral tags, so the text no longer shows who it is about
Mixture of experts	A model design that uses only a small part of the model for each word, which saves energy

Term	In plain words
MLOps	The tools and routines that train, test, release and monitor models
Model card	A public fact sheet about a model: what it learned from, how it was tested and where it falls short
Model registry	A catalogue of every model version, with its tests and approvals
Open model	A model whose trained parameters are published, so others can run it and train it further, within its licence
Parameters (B)	The numbers a model learns in training; 9B means 9 billion. Bigger models are usually more capable but cost more to run
Penetration test	Security experts try to break in on purpose, so holes are found and fixed
Pilot	A small trial with real users before a full launch
Privacy impact assessment	A formal check, under the Data Privacy Act, of how personal data is used and protected
PUE	Total data centre power divided by computing power; 1.5 means 50% extra for cooling and overhead
Rack unit	A standard slot in a server rack, about 4.4 cm high
Red teaming	Trying to break the AI on purpose, to find weak spots before customers do
Rollback	Switching back to the previous version quickly if a new one misbehaves
Router	The part that sends each question to the right-sized model
Runbook	Step-by-step instructions for running or fixing one part of the service
Safety tuning	Training that teaches a model to refuse harmful requests and follow Globe's rules
Scope 1 and 2	Direct emissions, and emissions from purchased electricity

Term	In plain words
Search index	A store of Globe documents arranged so the AI can find the right passage quickly
Serving	Running a trained model so it answers questions, as opposed to training it
Standby site	A smaller second set-up in another data centre that takes over if the main site fails
Surveillance audit	The yearly check a certification body makes to confirm a company still meets a standard
Synthetic data	Examples drafted by an AI rather than written by people; checked by hand before use
Taglish	The everyday mix of Tagalog and English heard in Filipino chats and calls
tCO2e	Tonnes of carbon dioxide equivalent, the standard unit for greenhouse gases
Telco	A telecoms company, such as Globe
Term sheet	A short summary of the main terms of a future contract, agreed before the full contract is written
Token	A small piece of text, often part of a word; models read and write in tokens
Training	Teaching a model with examples so it gets better at a language or a job

Names

Name	Who or what it is
ACEN	The Ayala group's energy company; it is moving over 3,000 Globe sites to renewable power by 2028
AI Singapore	Singapore's national AI research and talent programme, which builds the SEA-LION models
AIDE	Globe's in-house AI group, set up in 2024
Ayala	One of the largest business groups in the Philippines, and a major shareholder of Globe

Name	Who or what it is
Bangko Sentral ng Pilipinas	The central bank of the Philippines, which also regulates banks
CoreWeave and Nebius	GPU cloud companies whose public prices set the plan's GPU rates
Data Privacy Act	The Philippine law of 2012 (RA 10173) that protects personal data
DICT and DOST	The Philippine government departments for information and communications technology, and for science and technology
FilBench	A public test, published in 2025, of how well AI models handle Filipino
GlobeOne	Globe's app for customers to manage their accounts
Integrated Report	Globe's yearly report that brings together its financial and sustainability results
ITU-T L.1801	An international telecoms standard, from 2026, for measuring the environmental impact of AI
Llama and Gemma	Families of open AI models from Meta and Google
ML.ENERGY	A research project that measures how much energy AI models use
MosaicML Composer	Open software for training AI models
National Privacy Commission	The Philippine privacy regulator, which enforces the Data Privacy Act
NVIDIA NeMo	NVIDIA's open software for training large language models
OWASP	A non-profit that publishes widely used security checklists, including one for AI applications
Sailor2	A family of open models for Southeast Asian languages; its 2025 report counted how much text each language has online
SCI for AI	The Green Software Foundation's method for measuring the carbon of an AI system

Name	Who or what it is
SEA-HELM	AI Singapore's test suite for Southeast Asian languages
SEA-LION	AI Singapore's open models for Southeast Asian languages, including Filipino
SIM Registration Act	The 2022 law (RA 11934) that requires SIM cards to be registered and protects that data
Singtel	Singapore's telecoms group, a major shareholder of Globe
STT GDC Philippines	The data centre company Globe co-owns: Globe 50%, ST Telemedia Global Data Centres of Singapore 40%, Ayala 10%
vLLM and TensorRT-LLM	Open software for serving language models quickly and efficiently

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